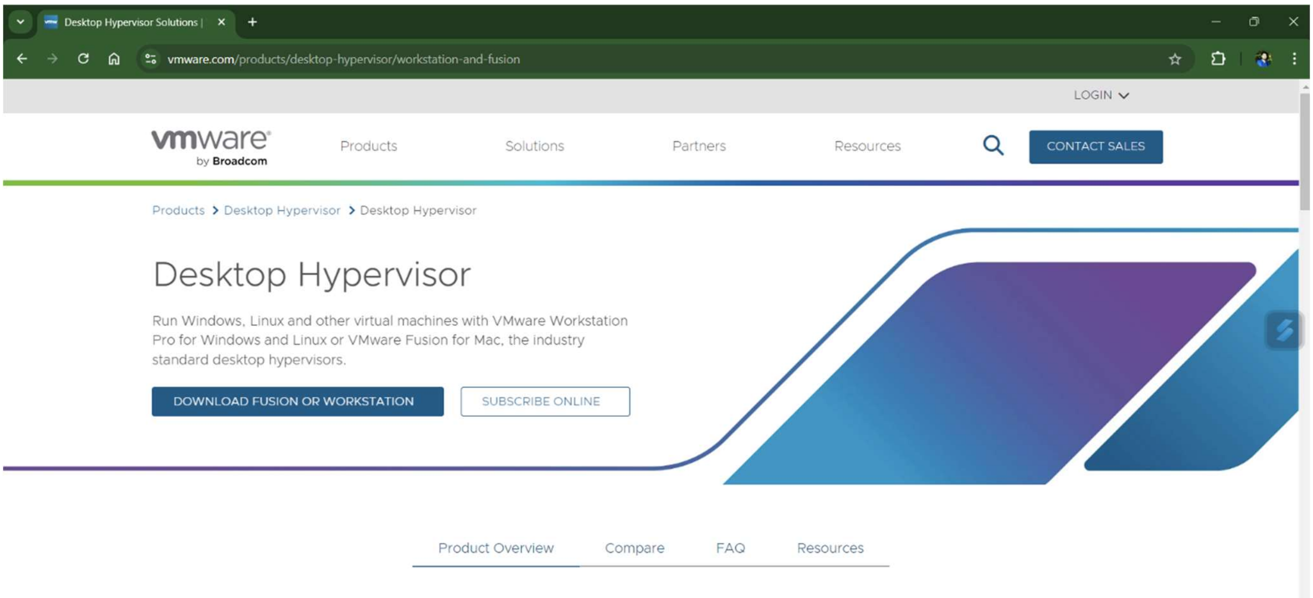
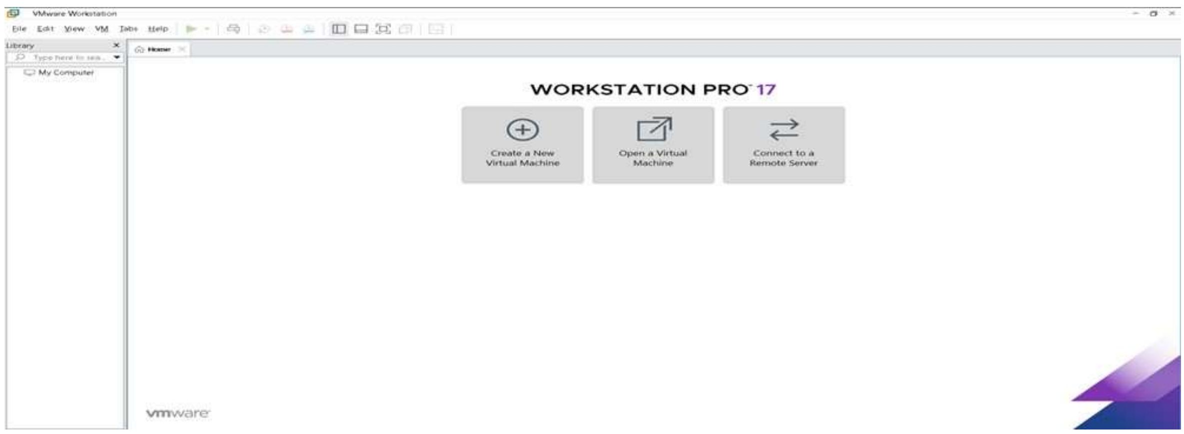
PRACTICAL:1

AIM: Create type 2 virtualization in VMware or any equivalent opensource Tool. Allocate memory and Storage space as per requirement. Install Guest OS on that VMWARE.

Step 1: Download and Install VMware Workstation Player from its official website.



Step 2: Create a New Virtual Machine, Open VMware Workstation Player, Click on "Create a New Virtual Machine".

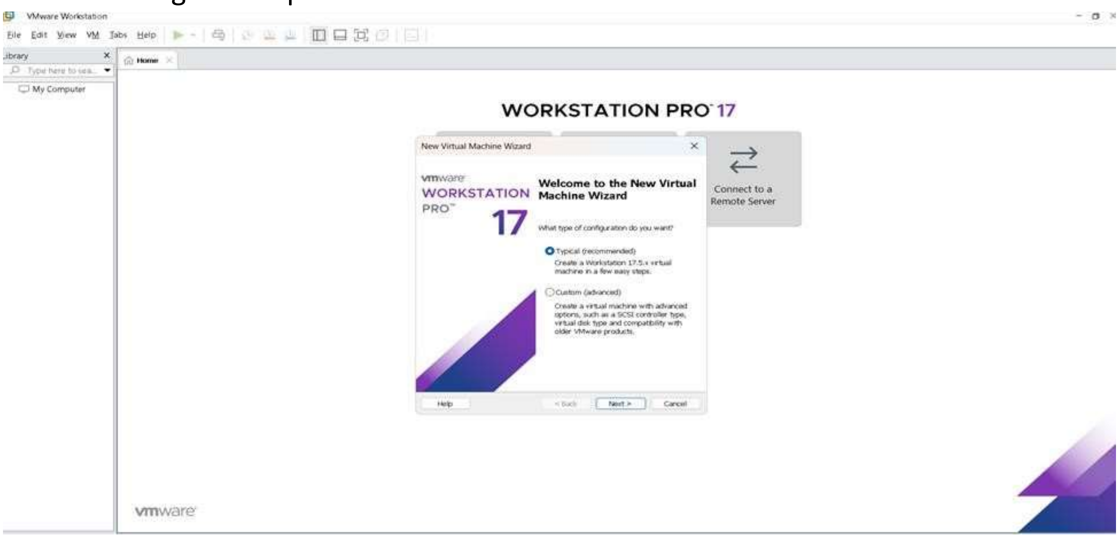


Step 3: Virtual Machine Configuration Wizard

The Virtual machine configuration Wizard will appear. Choose “Typical” configuration and click “Next”.

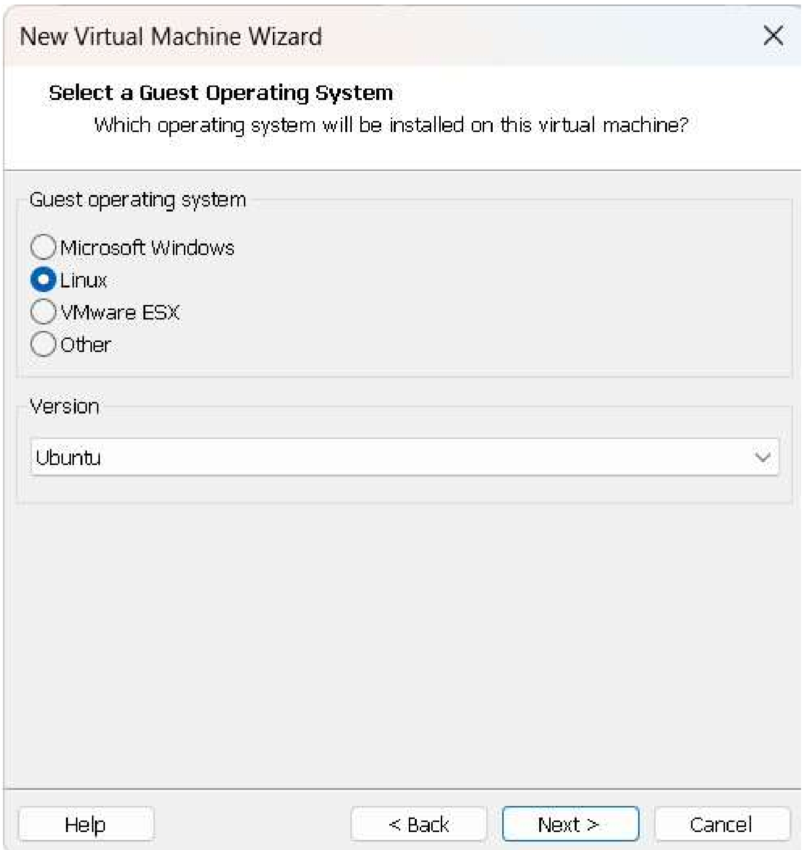
Step 4: Install Guest Operating System

- Choose the installation method for the guest OS. You can either install from a disc or image file (ISO) or choose to install later. If you have the ISO file for your guest OS, select it during this step . ▯ Click “Next”.



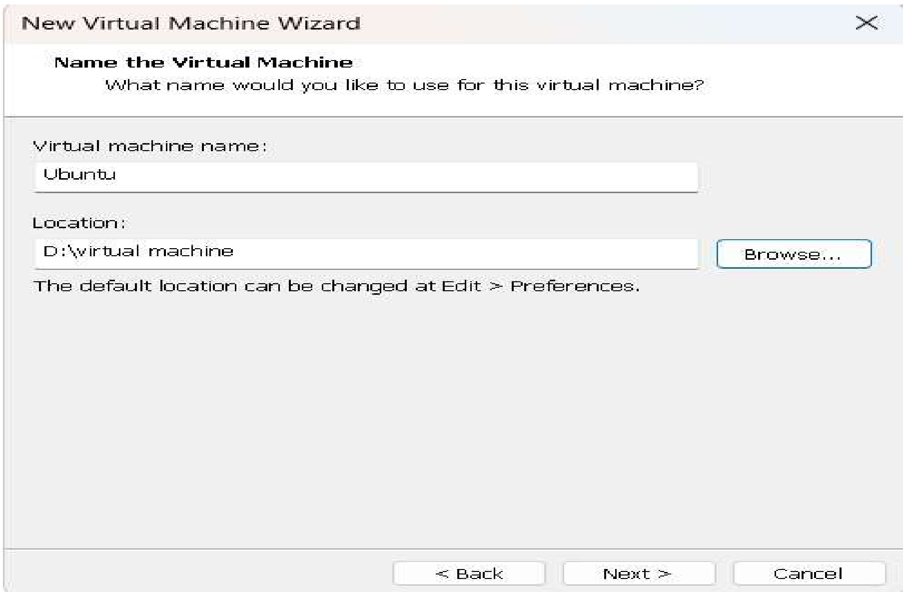
Step 5: Enter Guest OS Details

- Enter the name of your virtual machine and choose the location where you want to save it
- Select the appropriate guest operating system and version. For example, if you are installing Windows 10, choose "Windows" as the guest OS and "Windows 10 x64" as the version.
- Click "Next



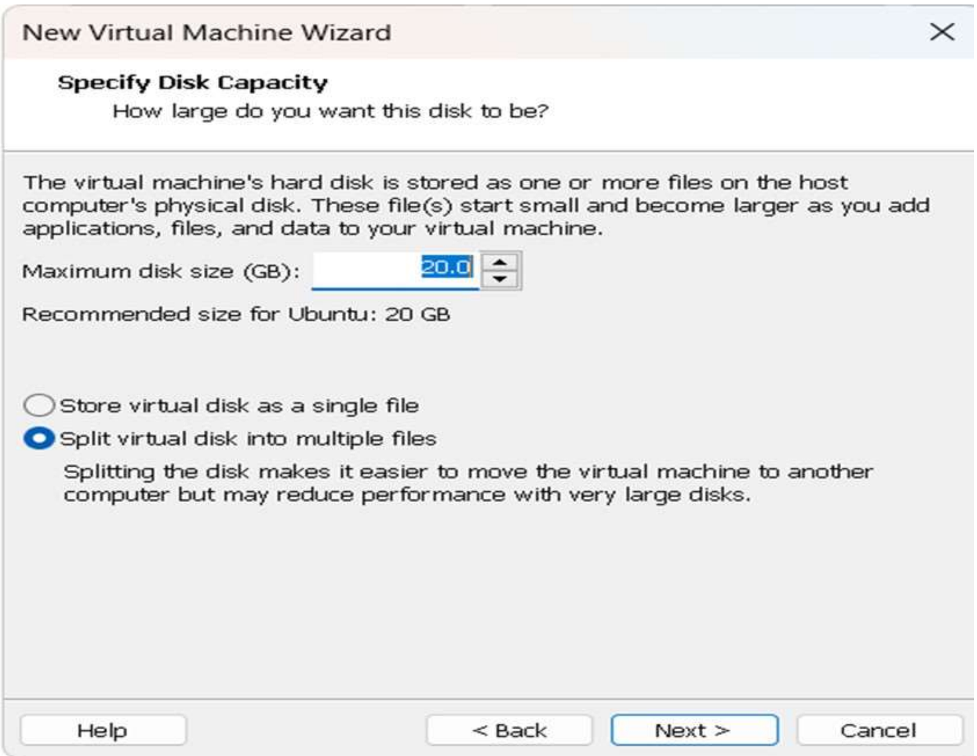
Step 6: Configure Virtual Machine Hardware

- Allocate memory: Choose how much RAM you want to allocate to the virtual machine. Make sure to leave enough memory for your host OS to run smoothly as well.
- Allocate storage: Choose whether to store the virtual disk as a single file or split into multiple files. Specify the disk size, and you can also choose to allocate all disk space now or let it grow as needed.
- Click "Next."



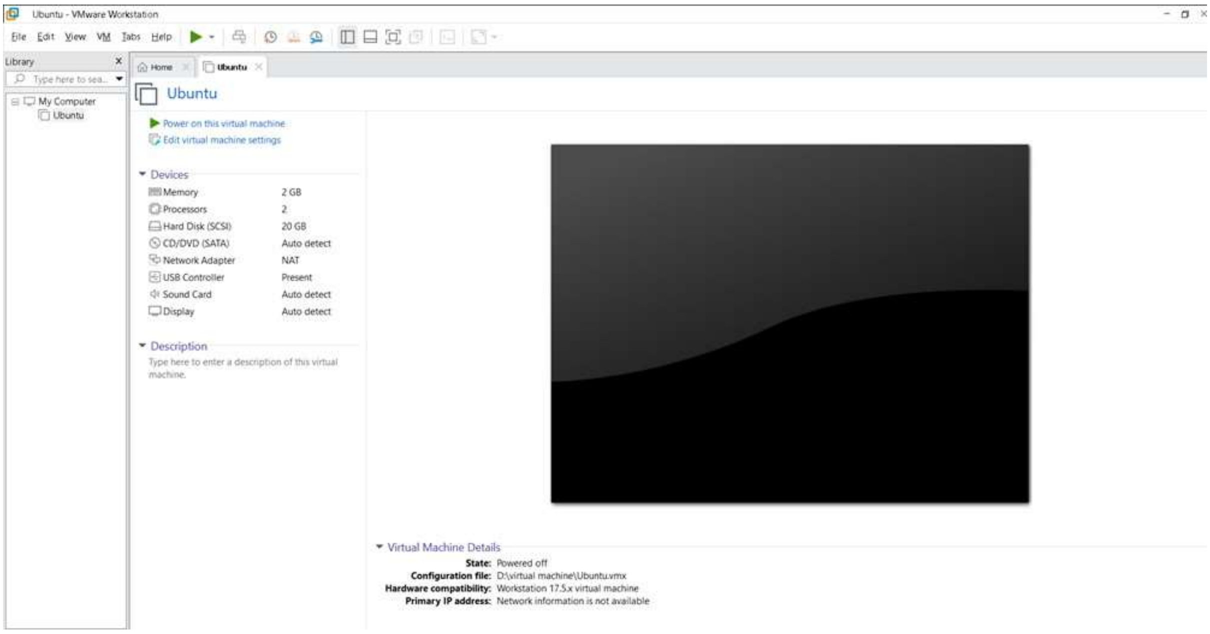
Step 7: Customize Hardware (Optional)

- If needed, you can customize the virtual machine's hardware settings like CPU cores, network adapters, graphics memory, etc. Otherwise, you can leave them as default.
- Click "Finish" once you are satisfied with the settings.



Step 8: Start Guest OS

- Start the virtual machine you just created. The virtual machine will boot from the ISO or installation media you provided earlier.
- Click on the “Power on this virtual machine”

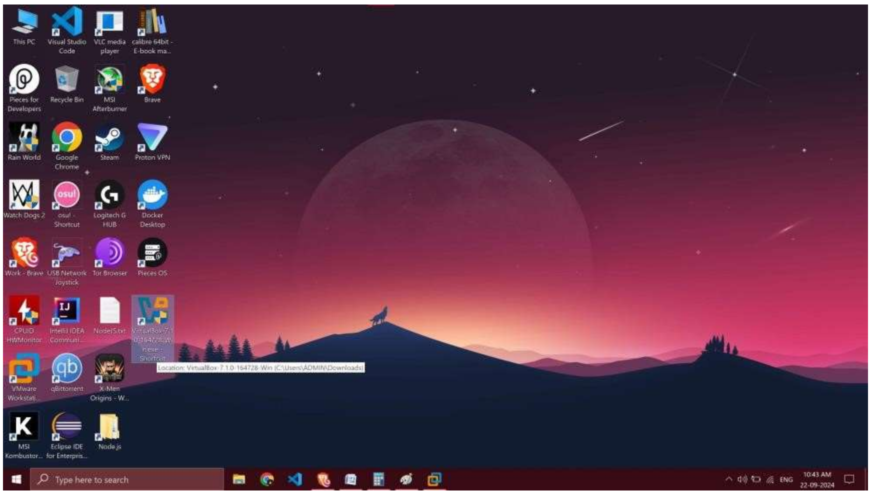


PRACTICAL : 2

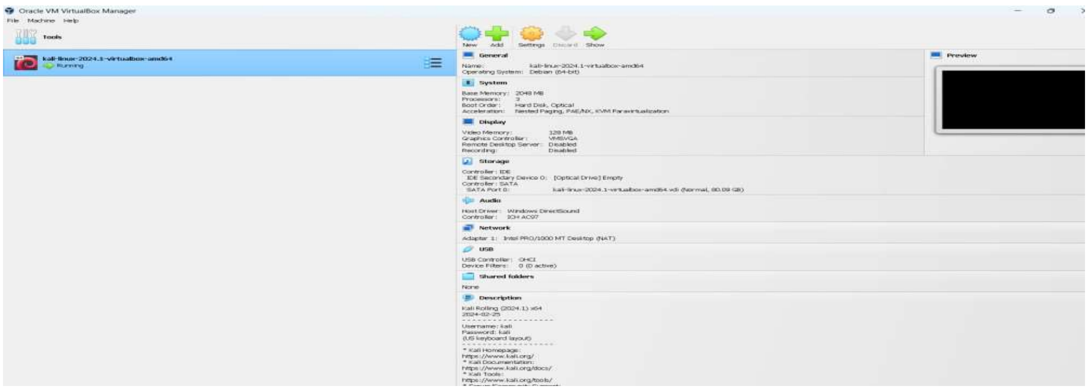
AIM: Launch VirtualBox and open the previously created VM. Explore settings and configurations for the VM (e.g., CPU, storage, display). Clone the VM to create a duplicate instance. Export and import VM appliances. Practice pausing, resuming, and shutting down the VM. Delete unnecessary snapshots and clones to optimize disk usage.

1. Launch VirtualBox and Open the Previously Created VM

1. **Launch VirtualBox:**
   - Open the VirtualBox application from your desktop or start menu.

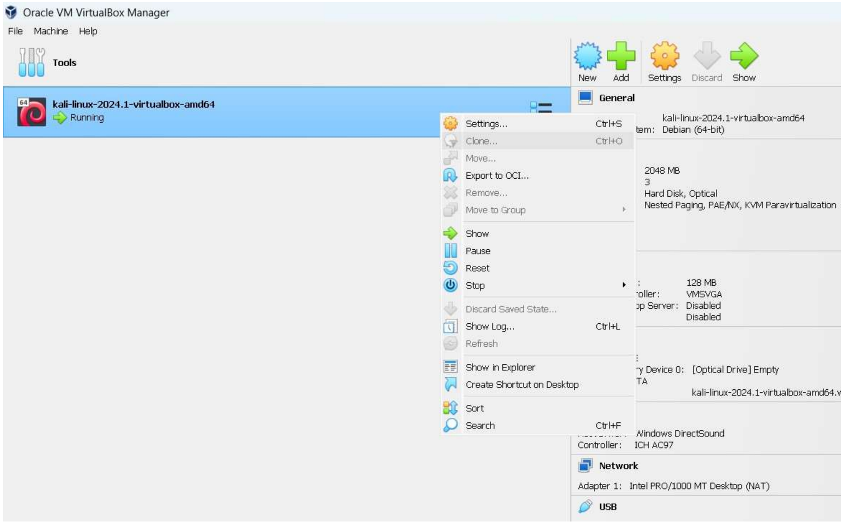


2. **Open the Previously Created VM:**
   - Click on the "Start" button to launch the VM.

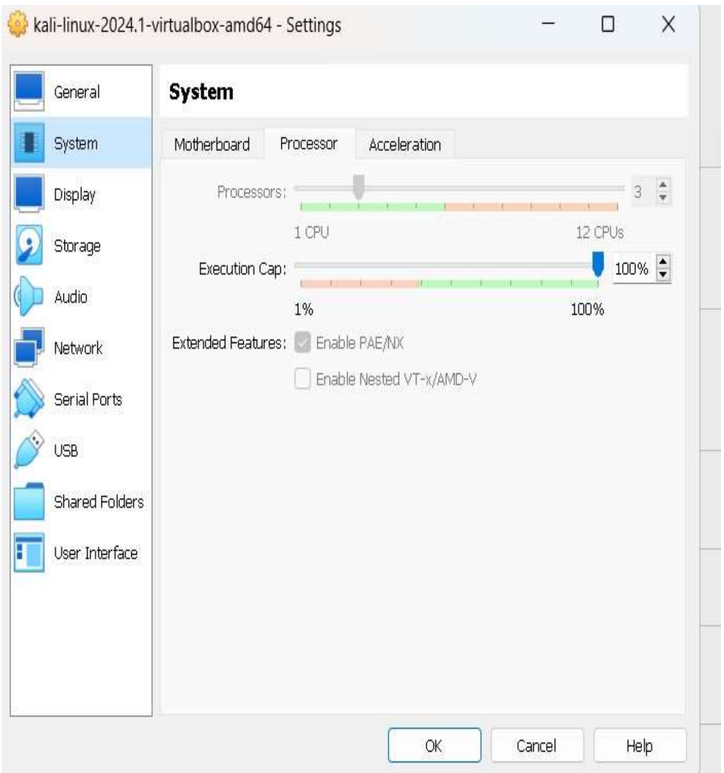


2. Explore Settings and Configurations for the VM

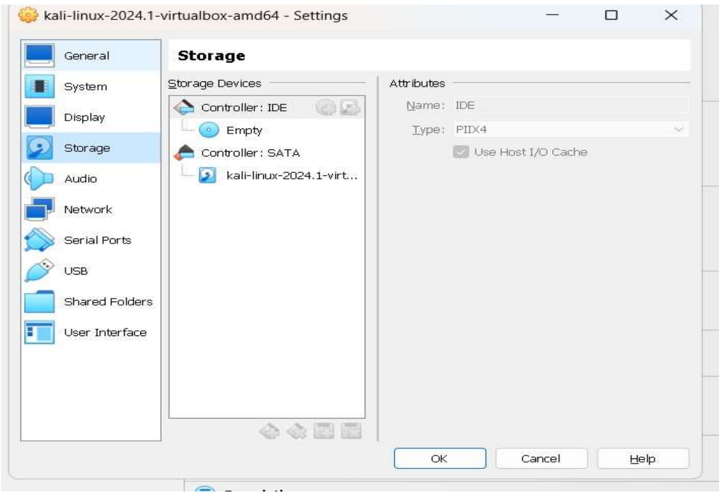
1. **Access VM Settings:**
   - Right-click on your Kali Linux VM and select "Settings".



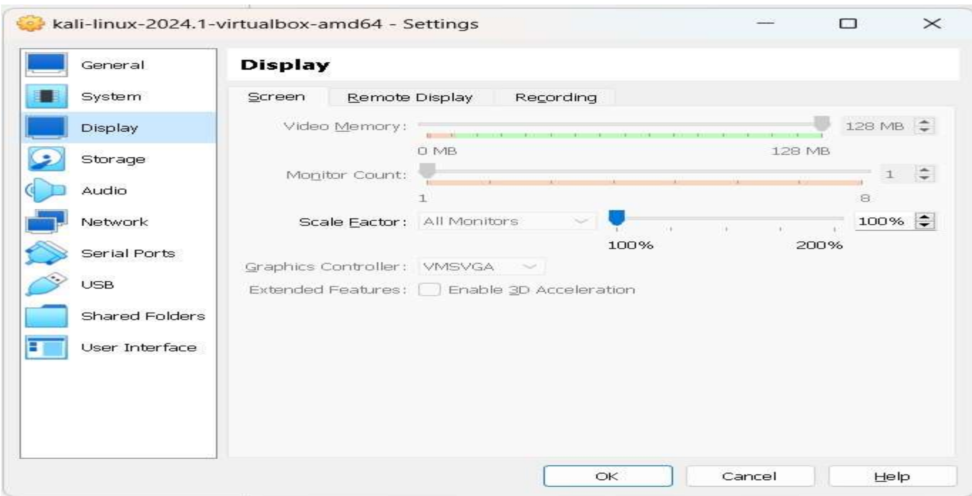
2. **CPU Configuration:**
   - Go to the "System" tab, then the "Processor" tab.
   - Adjust the number of CPU cores assigned to the VM.



3. **Storage Configuration:**
   - Go to the "Storage" tab. o Review and modify the attached storage devices.



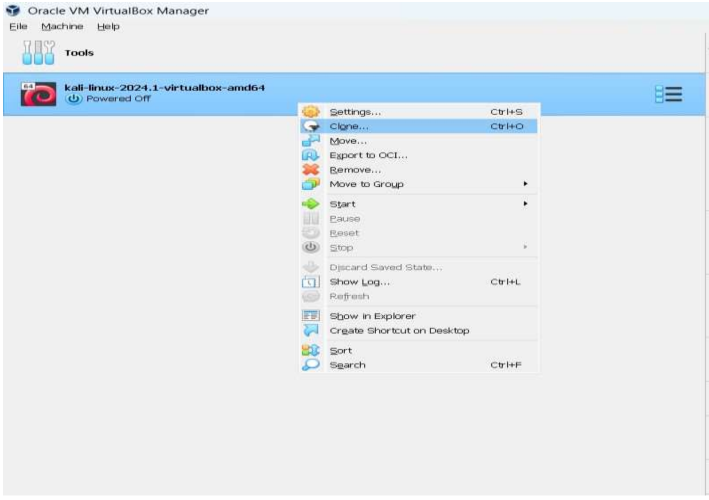
4. **Display Configuration:**
   - Go to the "Display" tab. o Adjust the video memory and other display settings.



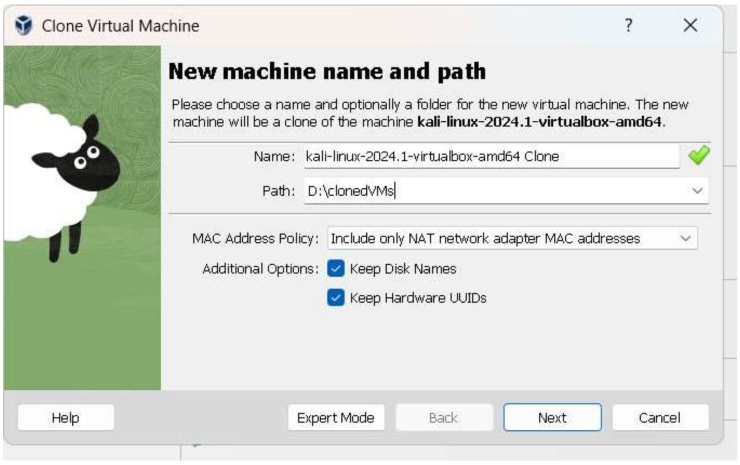
3. Clone the VM to Create a Duplicate Instance

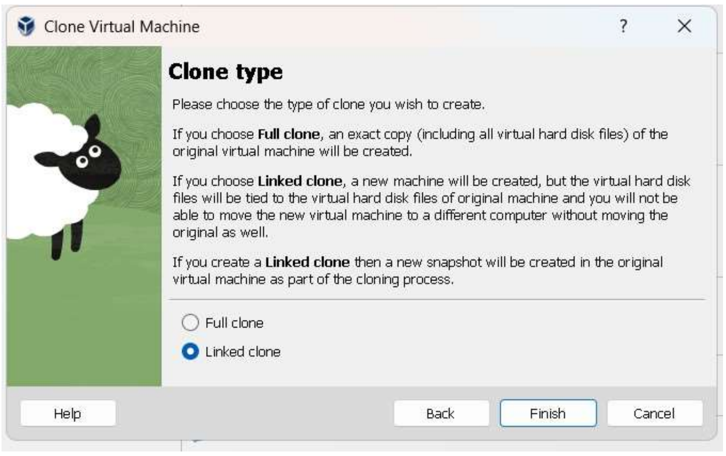
1. **Clone the VM:**

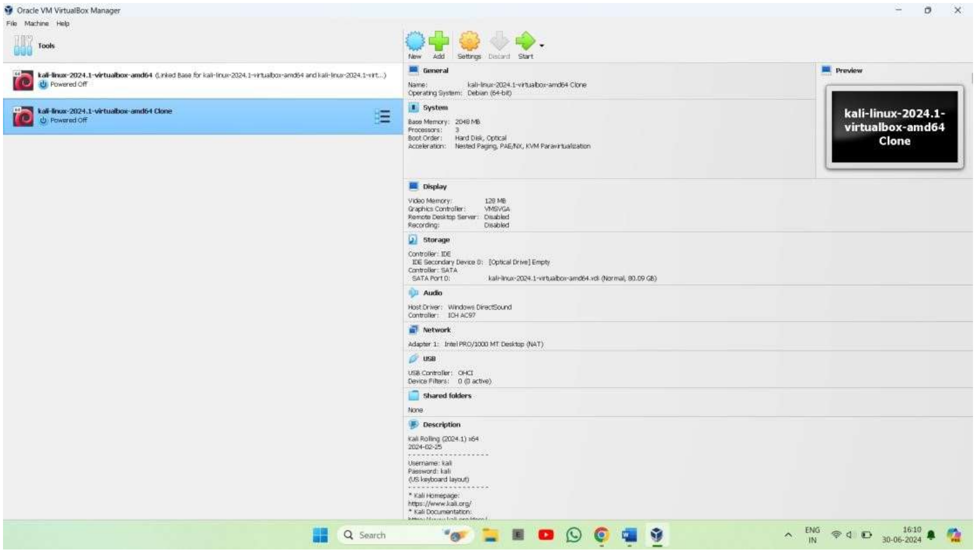
   - Right-click on your Kali Linux VM and select "Clone".



   - Follow the prompts to name the clone and select cloning options (full clone or linked clone).





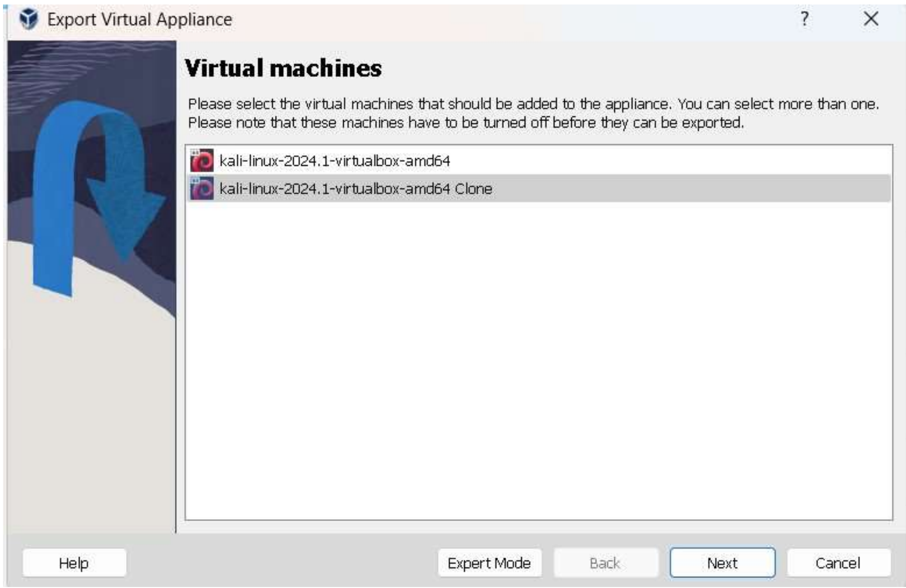


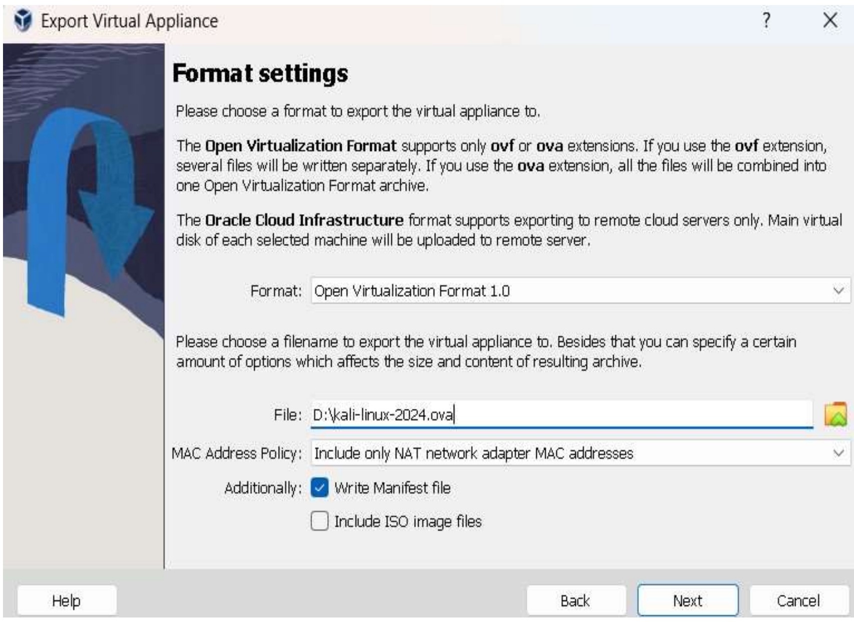
4. Export and Import VM Appliances

1. **Export VM Appliance:**

   - Go to "File" > "Export Appliance".

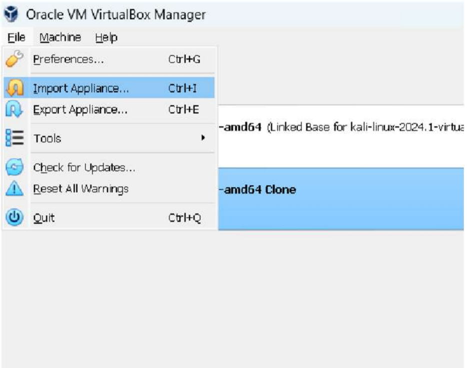
- Select the VM you want to export and follow the prompts to save it as an .ova file.



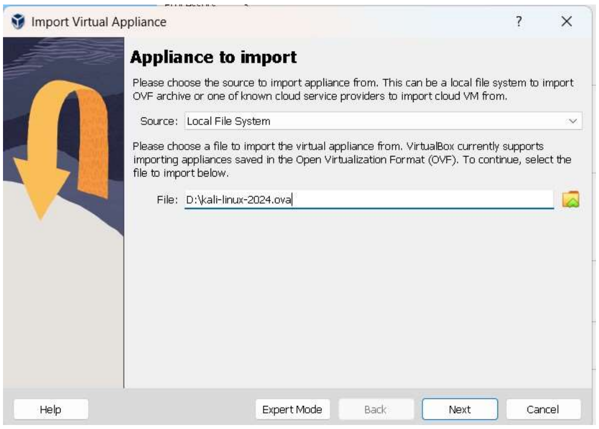


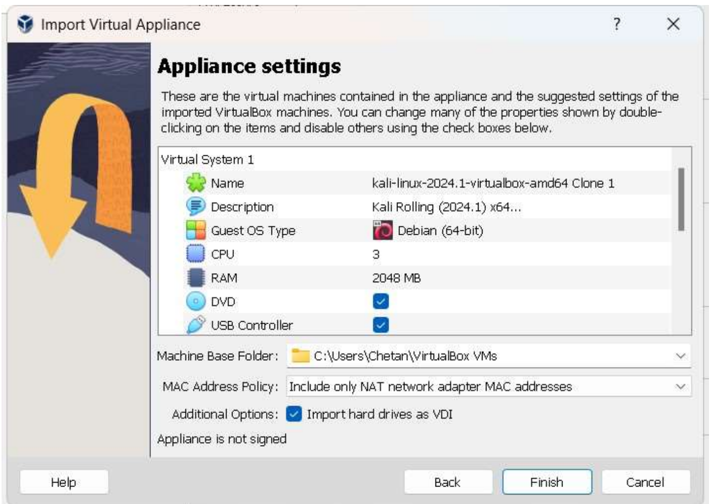
2. **Import VM Appliance:**

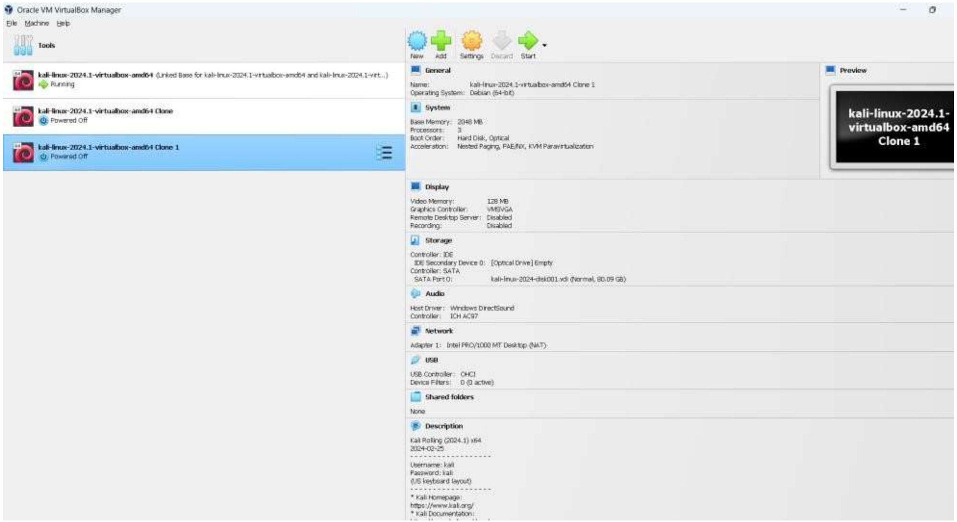
- Go to "File" > "Import Appliance".



- Select the .ova file you previously exported and follow the prompts to import it.



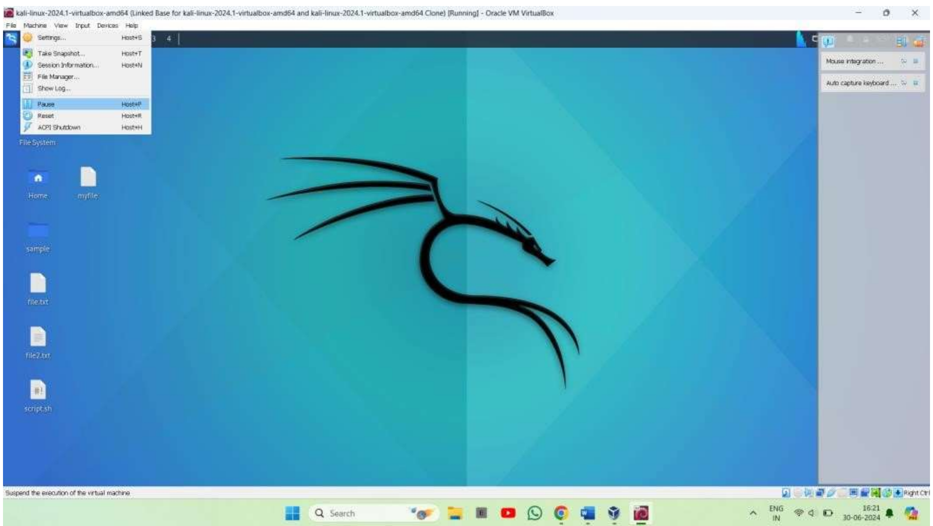




5. Practice Pausing, Resuming, and Shutting Down the VM

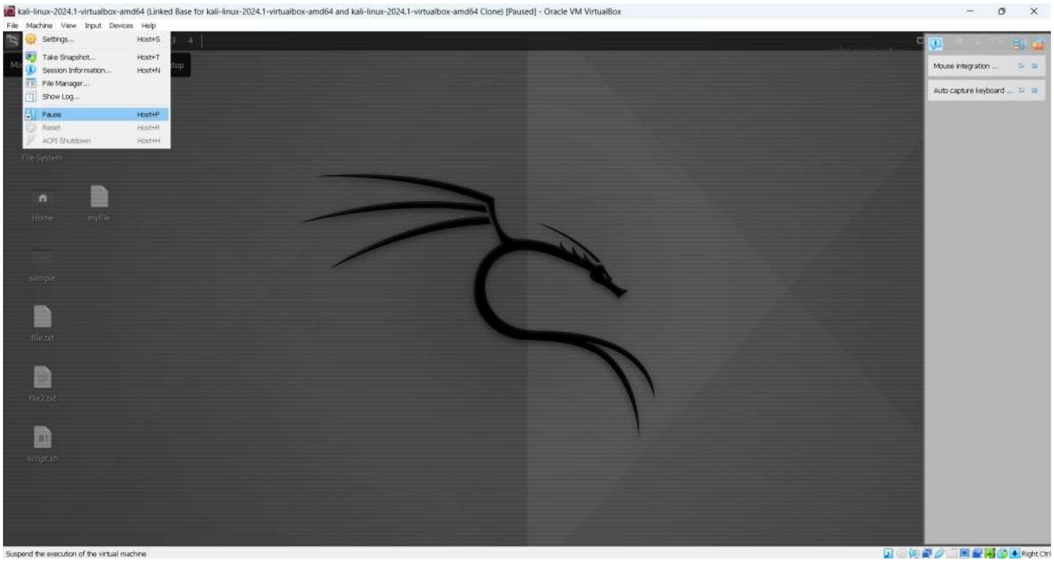
1. **Pause the VM:**

   - With the VM running, go to "Machine" > "Pause".



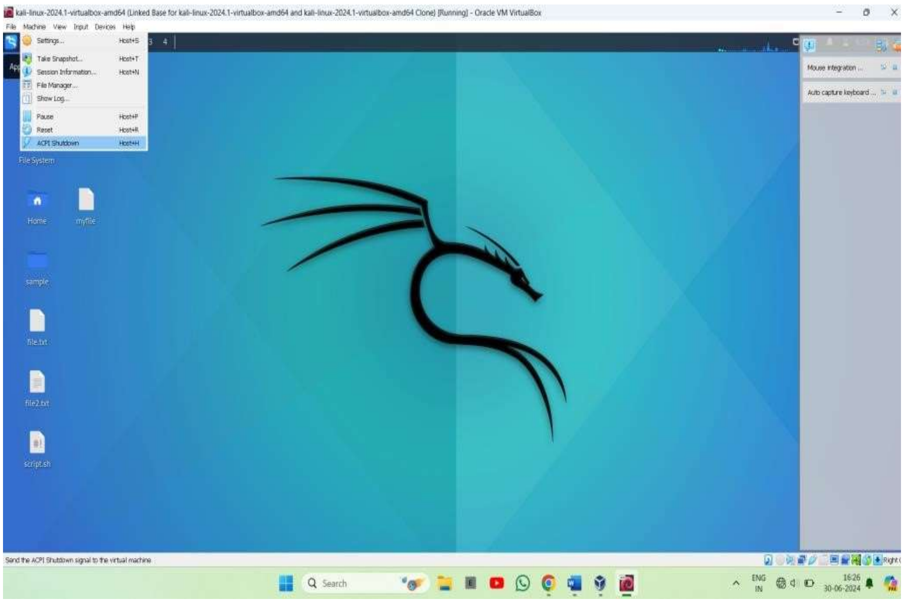
2. **Resume the VM:**

    - Go to "Machine" > Click the
      "Pause " button again to resume the VM.



3. **Shut Down the VM:**

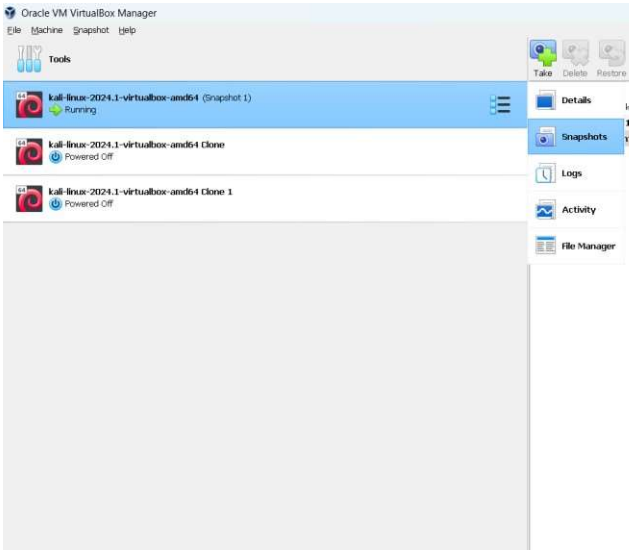
    - Inside the VM, use the OS shutdown option or go to "Machine" > "ACPI Shutdown"



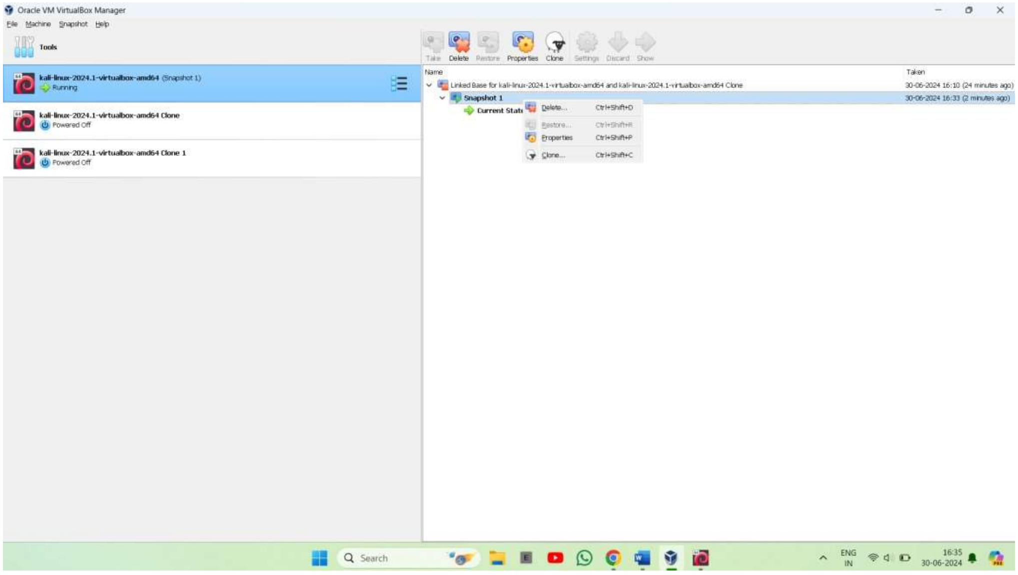
6. Delete Unnecessary Snapshots and Clones to Optimize Disk Usage

1. **Delete Snapshots:**

    - Right-click on your VM and select "Snapshots".

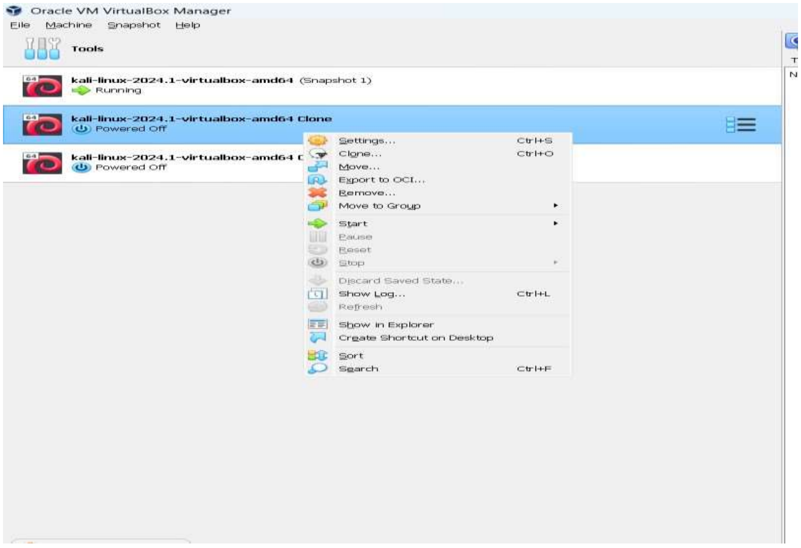


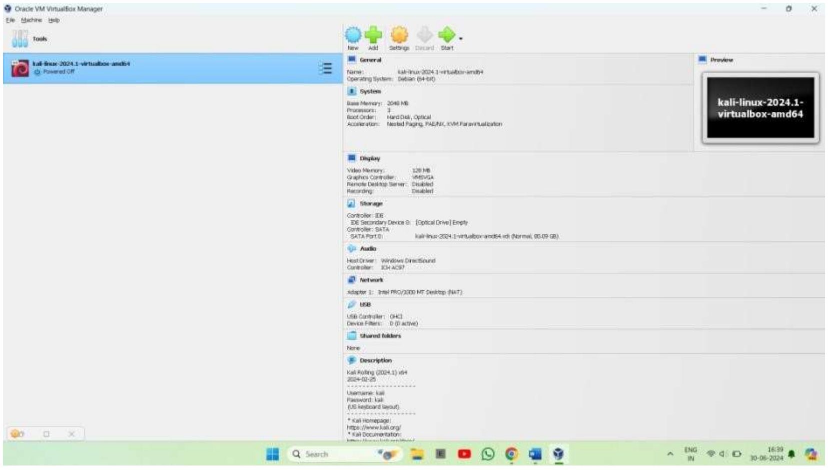
    - Select the snapshot you want to delete and click "Delete".



2. **Delete Clones:**

- Right-click on the clone VM and select "Remove".



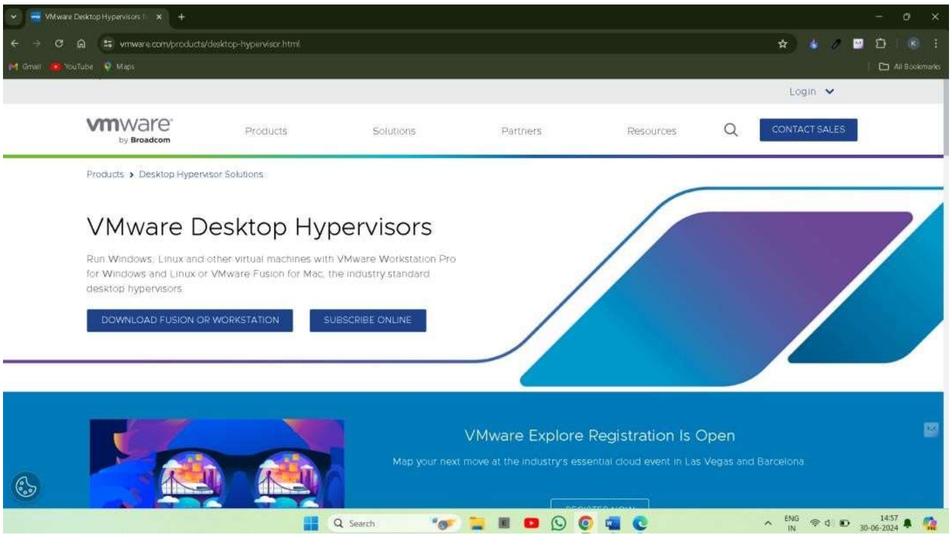


- Choose whether to delete all files associated with the clone.

PRACTICAL :3

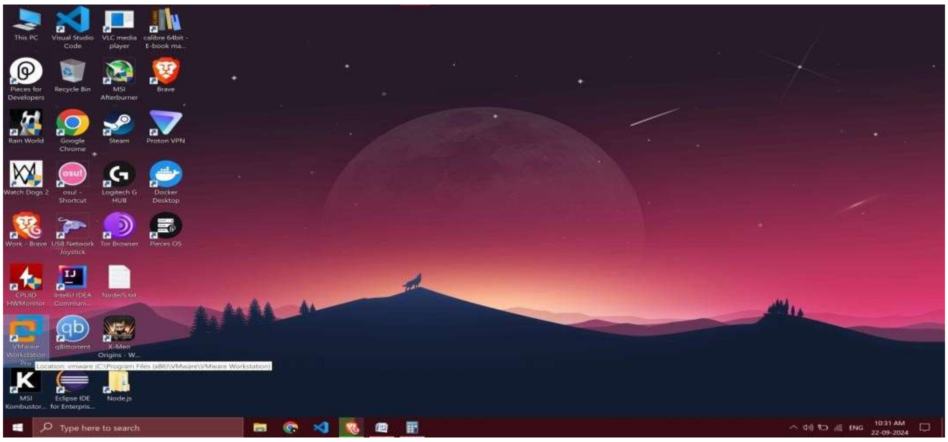
AIM: Install VMware Workstation on the host machine. Launch VMware Workstation and explore its interface. Create a new virtual machine using the New Virtual Machine Wizard.

Step 1: Visit the official VMware website and download the installer for VMware Workstation. Choose the version compatible with your operating system (Windows or Linux).

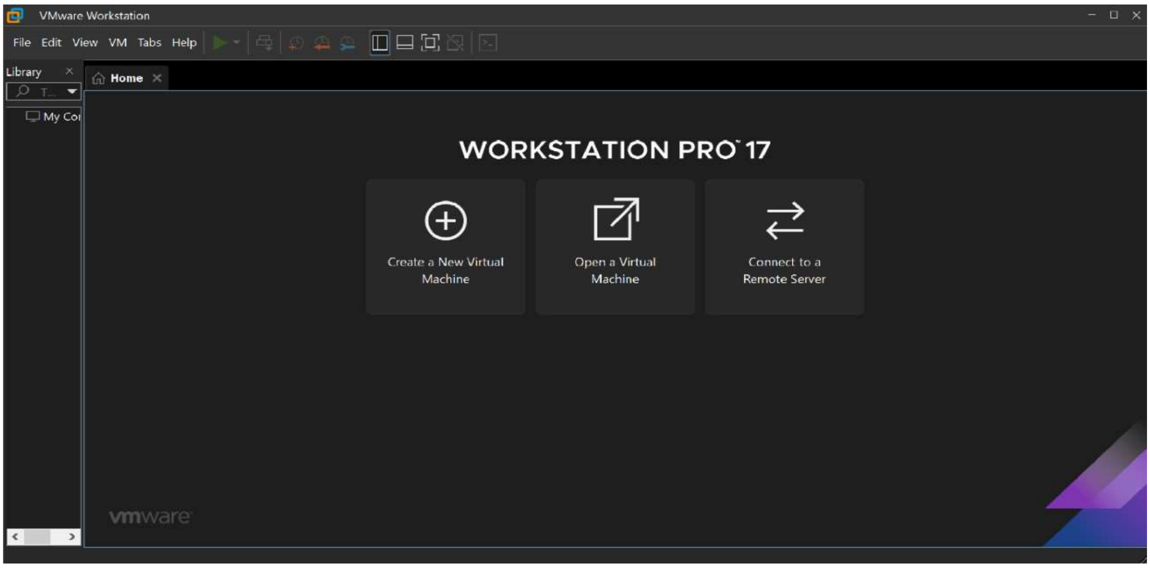


Step 2: Install VMware Workstation:

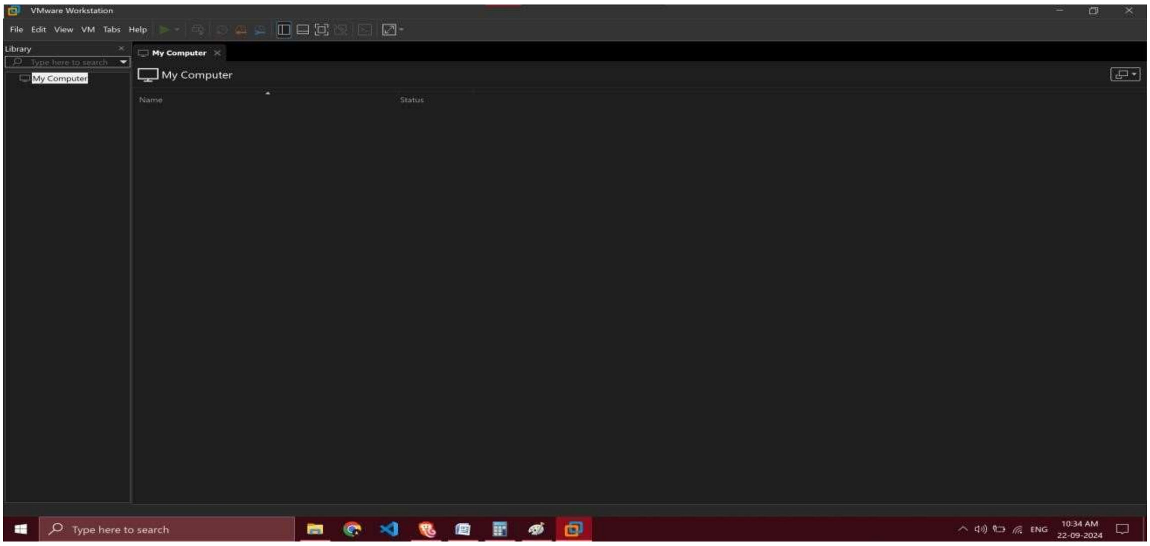
- Run the downloaded installer.



- Follow the on-screen instructions to complete the installation.



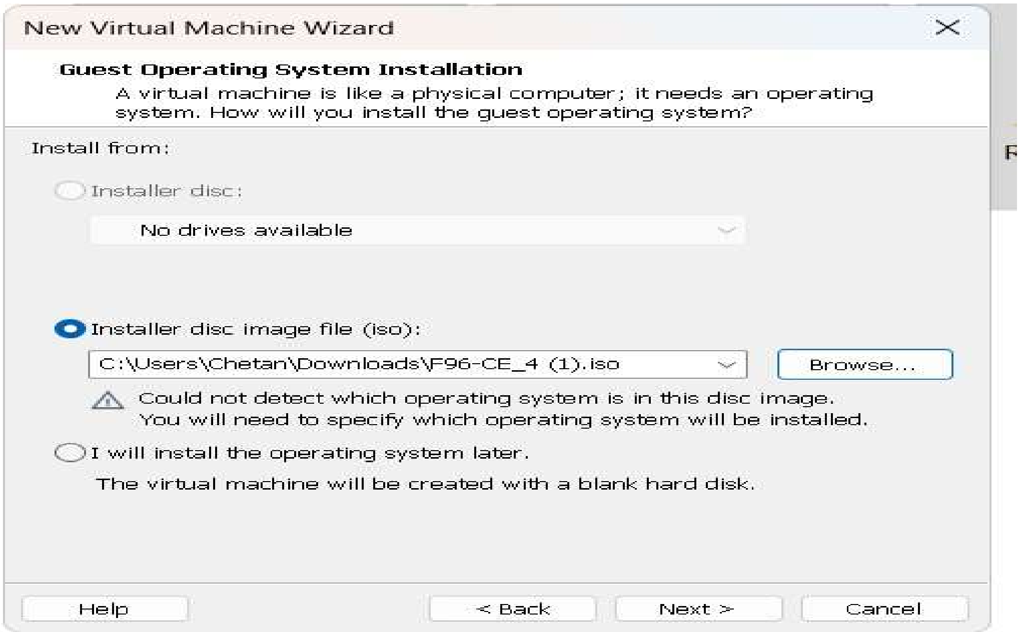
- Complete the installation and launch VMware Workstation



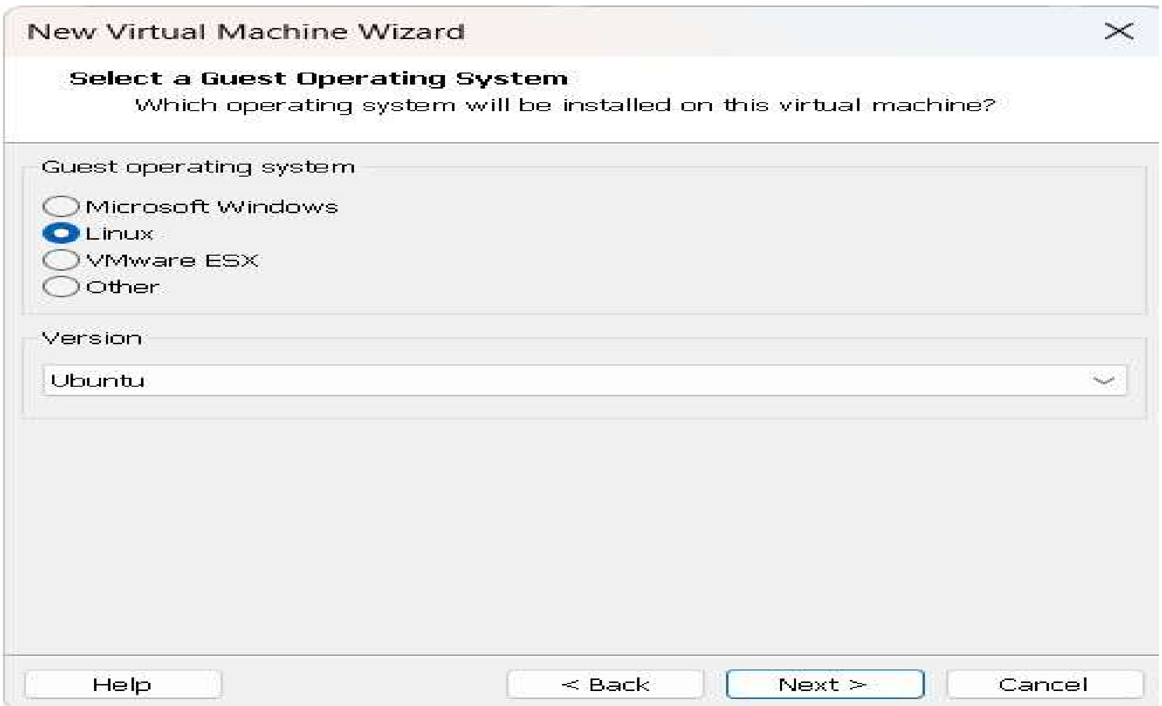
Create a New Virtual Machine

- Launch the New Virtual Machine Wizard:
  - Select Typical (recommended) and click Next.

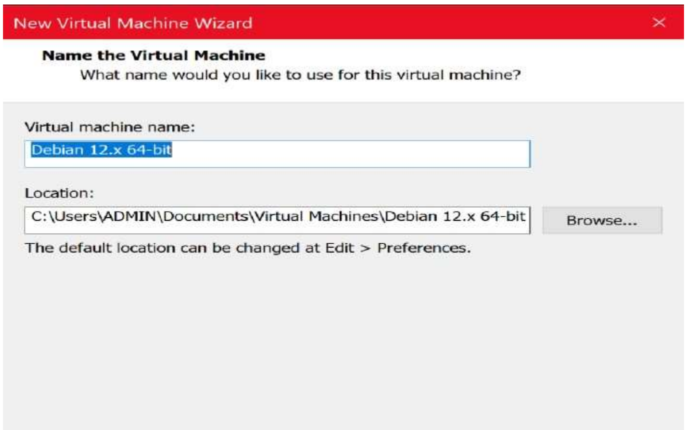
- Choose how you will install the operating system. You can either: ○ Use an installer disc. ○ Use an ISO image. ○ Install later.



- **Select Guest Operating System:** ○ Choose the operating system you plan to install (e.g., Linux). ○ Select the version and click Next.

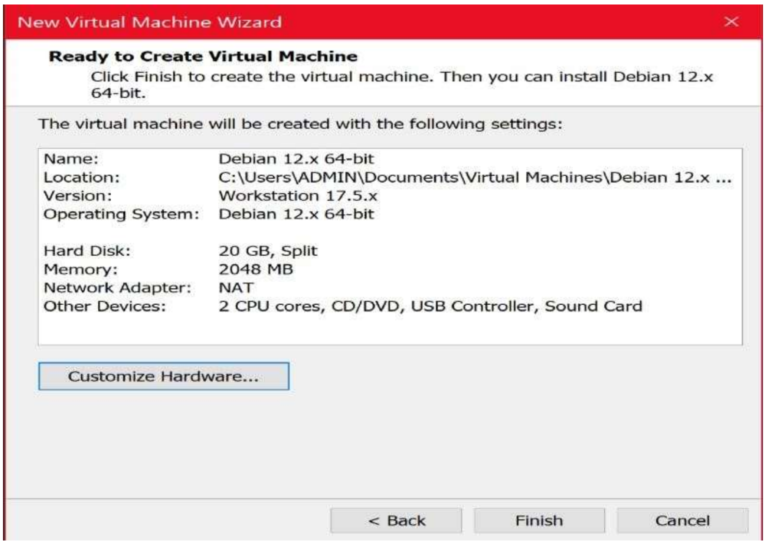


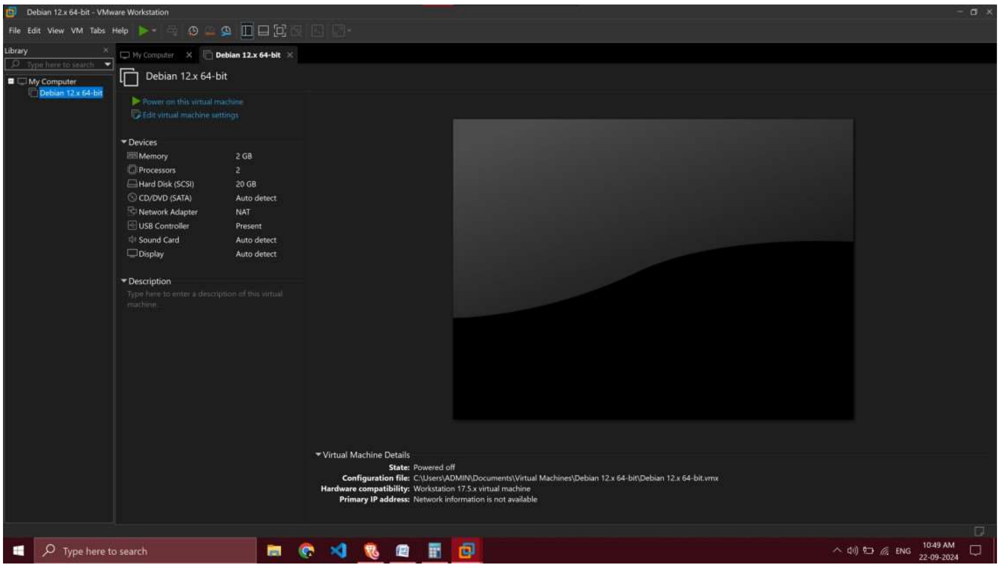
- **Name the Virtual Machine:**
    - Provide a name for your virtual machine.
    - Choose a location to store the virtual machine files.

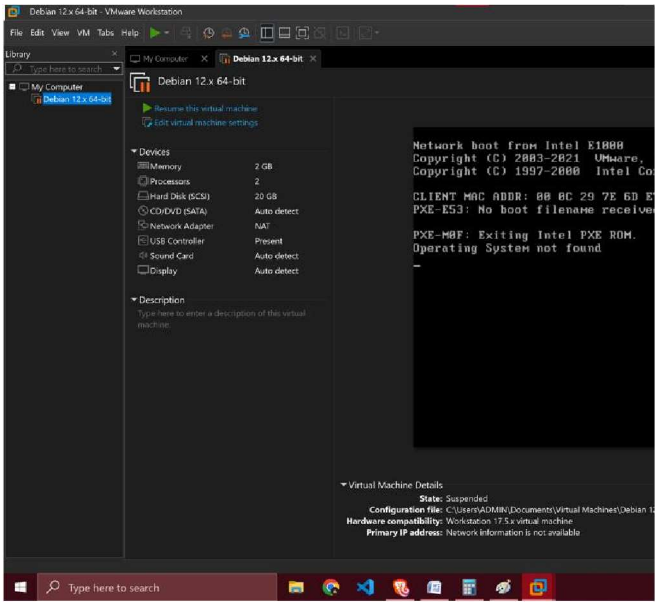


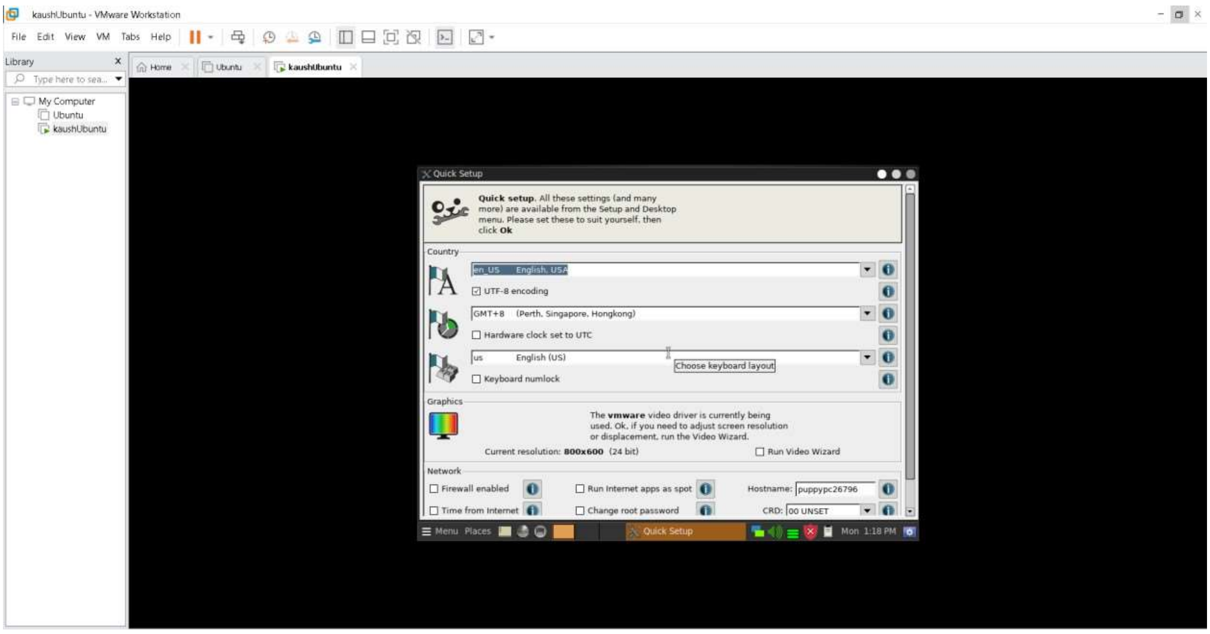
- **Specify Disk Capacity:**
    - Set the maximum disk size for the virtual machine.
    - Choose whether to store the virtual disk as a single file or split it into multiple files.

- **Finish the Wizard:**
    - Review your settings and click Finish to create the virtual machine


New Virtual Machine Wizard
Ready to Create Virtual Machine
Click Finish to create the virtual machine. Then you can install Debian 12.x 64-bit.
The virtual machine will be created with the following settings:
Name: Debian 12.x 64-bit
Location: C:\Users\ADMIN\Documents\Virtual Machines\Debian 12.x ...
Version: Workstation 17.5.x
Operating System: Debian 12.x 64-bit
Hard Disk: 20 GB, Split
Memory: 2048 MB
Network Adapter: NAT
Other Devices: 2 CPU cores, CD/DVD, USB Controller, Sound Card
Customize Hardware...
< Back
Finish
Cancel


Debian 12.x 64-bit


Debian 12.x 64-bit - VMware Workstation
Debian 12.x 64-bit
Network boot from Intel E1000
Copyright (C) 2003-2021 VMware,
Copyright (C) 1997-2000 Intel Co
CLIENT MAC ADDR: 00 0C 29 7E 6D E
PXE-E53: No boot filename receive
PXE-M0F: Exiting Intel PXE ROM.
Operating System not found


kaushUbuntu - VMware Workstation
Quick Setup
Quick setup. All these settings (and many more) are available from the Setup and Desktop menu. Please set these to suit yourself, then click Ok
Country
en_US English, USA
UTF-8 encoding
GMT+8 (Perth, Singapore, Hongkong)
Hardware clock set to UTC
us English (US)
Choose keyboard layout
Keyboard numlock
Graphics
The vmware video driver is currently being used. Ok, if you need to adjust screen resolution or displacement, run the Video Wizard.
Current resolution: 800x600 (24 bit)
Run Video Wizard
Network
Firewall enabled
Run Internet apps as spot
Hostname: puppypc26796
Time from Internet
Change root password

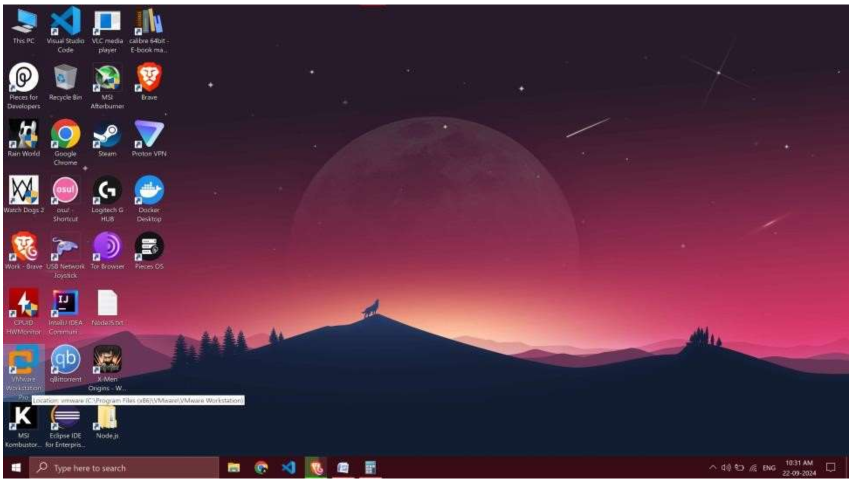
PRACTICAL :4

AIM: Open VMware setup in previous Lab. Customise VM settings (e.g., hardware compatibility, disk size, network adapter). Install an operating system on the VM using an ISO image.

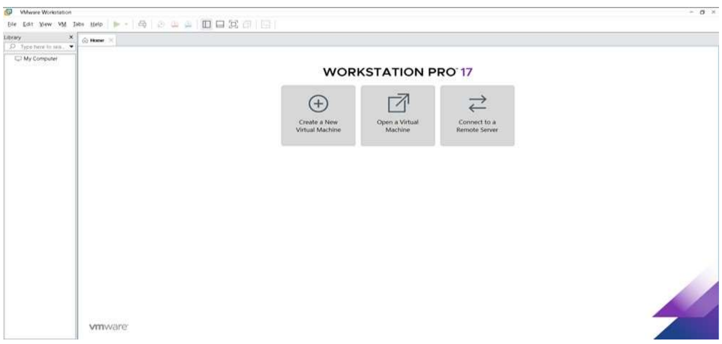
Step 1: Open VMware Workstation

Launch VMware Workstation:

- Double-click the VMware Workstation icon on your desktop or find it in your start menu and click to open.



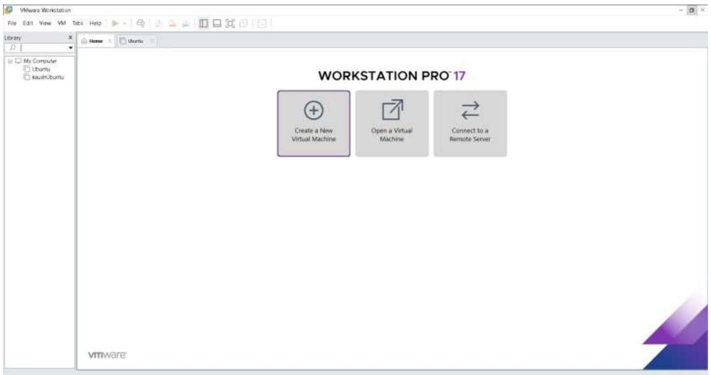
- The VMware Workstation interface will open, displaying the main dashboard.



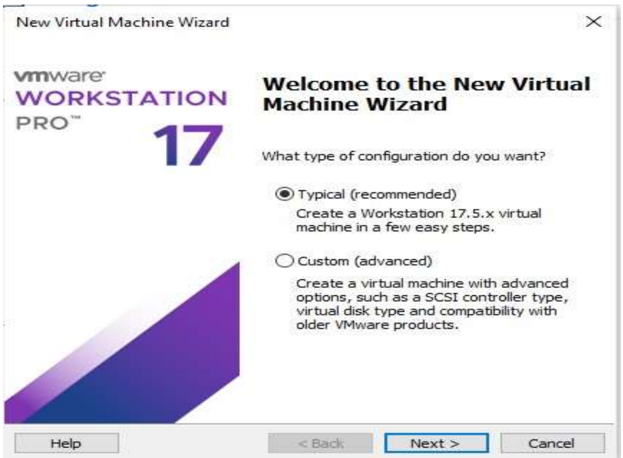
Step 2: Create a New Virtual Machine

Create a New VM:

- Click on File > New Virtual Machine or use the Create a New Virtual Machine option on the home screen.



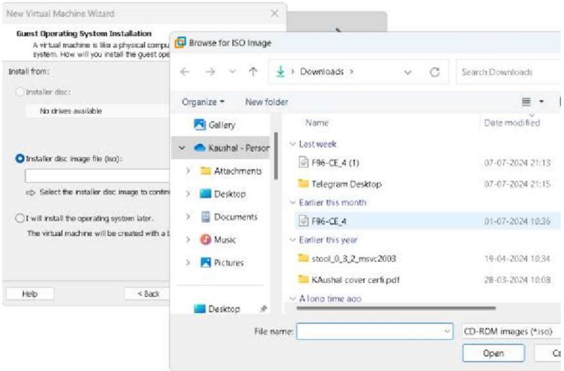
- The New Virtual Machine Wizard will appear. Select Typical (recommended) for the setup type and click Next.



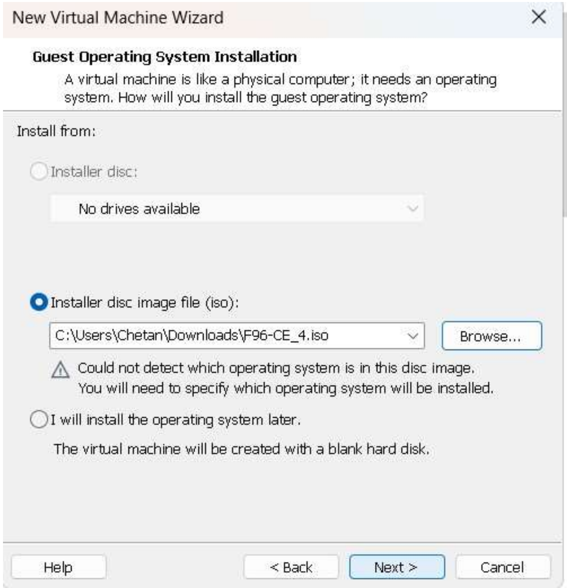
Step 3: Customize VM Settings

Select Installation Media:

- Choose Installer disc image file (iso) and click Browse to locate the ISO file on your system.

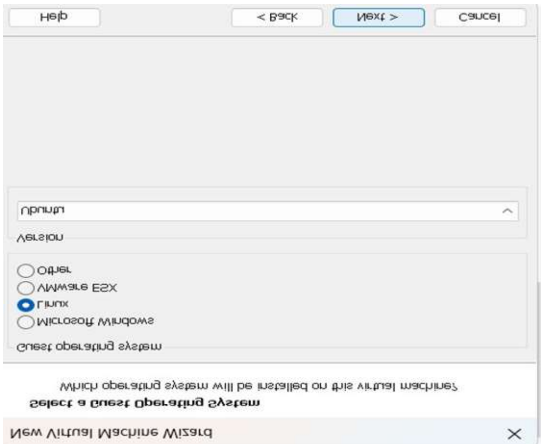


- After selecting the ISO file, click Next.



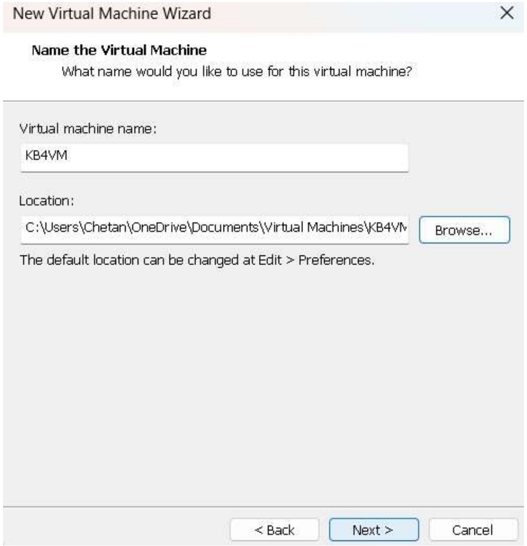
Guest Operating System:

- Choose the operating system you are going to install (e.g., Windows, Linux) and select the appropriate version from the dropdown. Click Next.



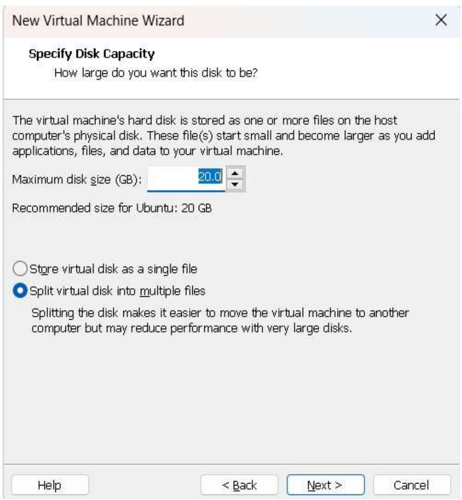
Name the Virtual Machine:

- Provide a name for your virtual machine. Choose a name that helps you identify the VM easily.
- Select a location on your disk where the VM files will be stored. Click Next.



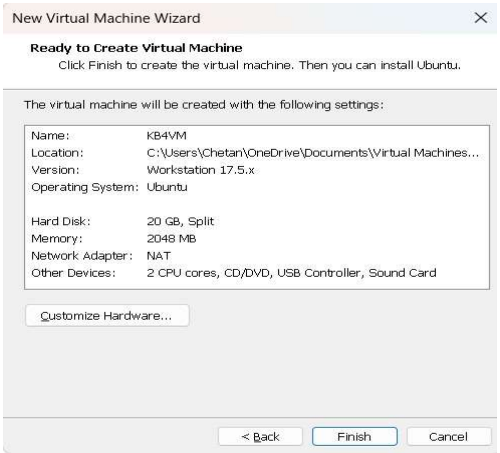
Specify Disk Capacity:

- Set the maximum disk size for your VM. The default is usually sufficient, but you can adjust it based on your needs.
- Decide whether to store the virtual disk as a single file or split it into multiple files. For better performance, it's often recommended to use a single file. Click Next.

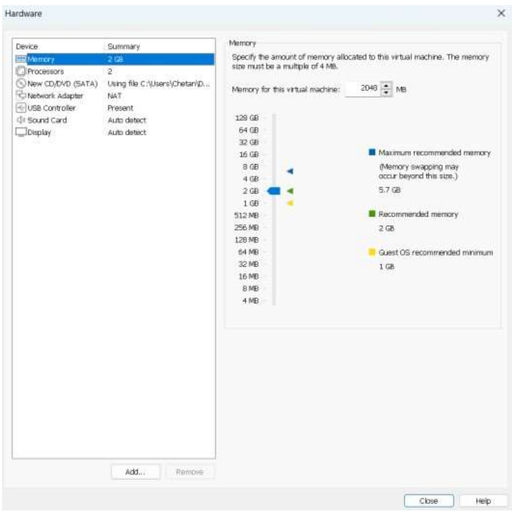


Customize Hardware:

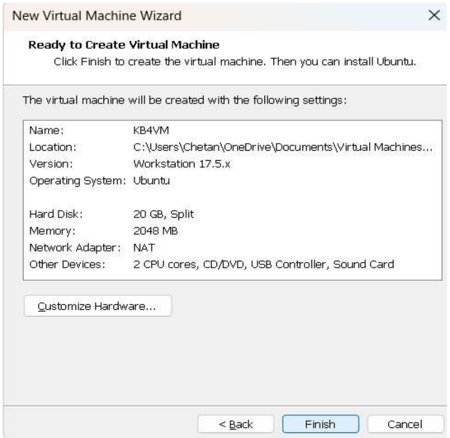
- Click on Customize Hardware to open the hardware settings window.



- Here, you can adjust various hardware settings such as: ○ **Memory:** Increase the RAM allocation for the VM. ○ **Processors:** Set the number of processor cores.
  - **Network Adapter:** Choose between Bridged, NAT, or Host-only network types.
  - **Hard Disk:** Adjust the disk size if necessary.
- Once you've customized the settings, click Close.



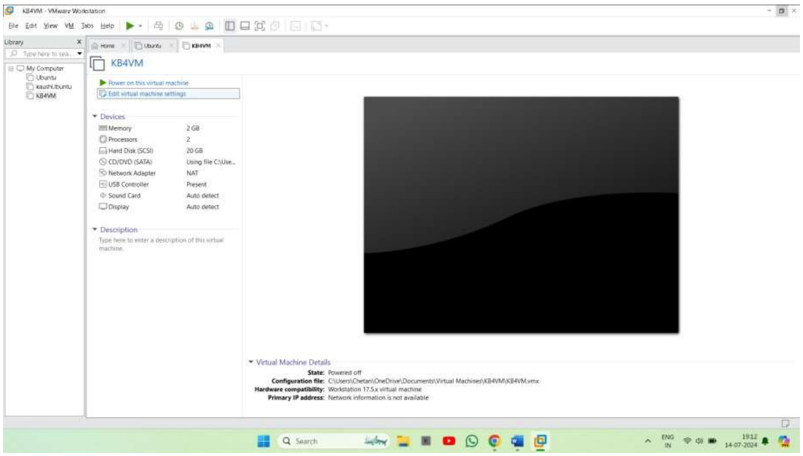
- Click Finish to complete the VM creation process.



Step 4: Attach ISO Image

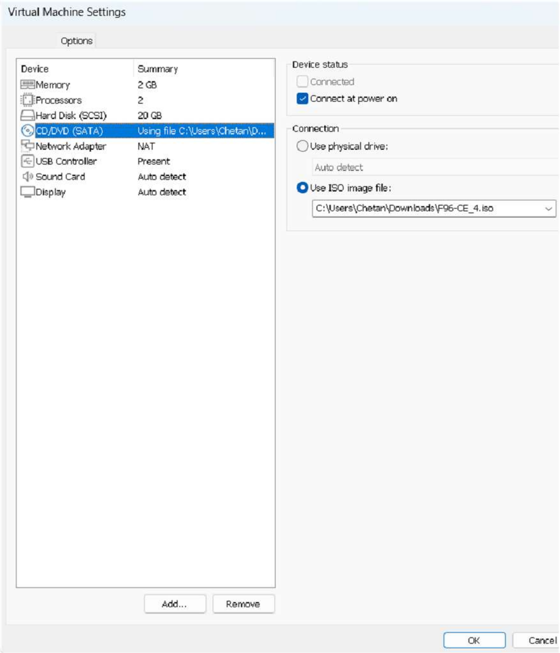
Edit Virtual Machine Settings:

- Select your newly created virtual machine from the list.
- Click on Edit virtual machine settings to open the settings window.



Attach ISO Image:

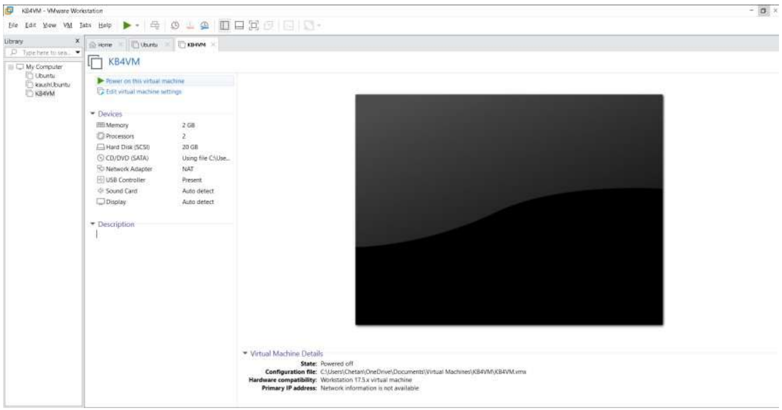
- In the virtual machine settings, go to CD/DVD (SATA) or CD/DVD (IDE) depending on what is available.
- Select Use ISO image file and click Browse to locate the ISO file on your system.
- Ensure the Connect at power on checkbox is selected so that the ISO is used when the VM starts.



Step 5: Power On the Virtual Machine

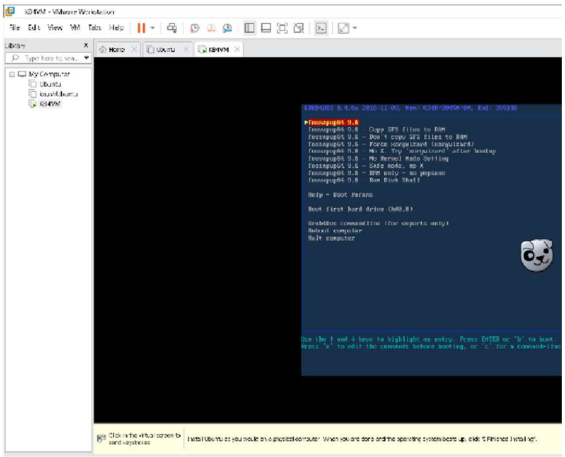
Power On:

- Select your virtual machine and click on Power on this virtual machine from the VM menu or use the play button icon.

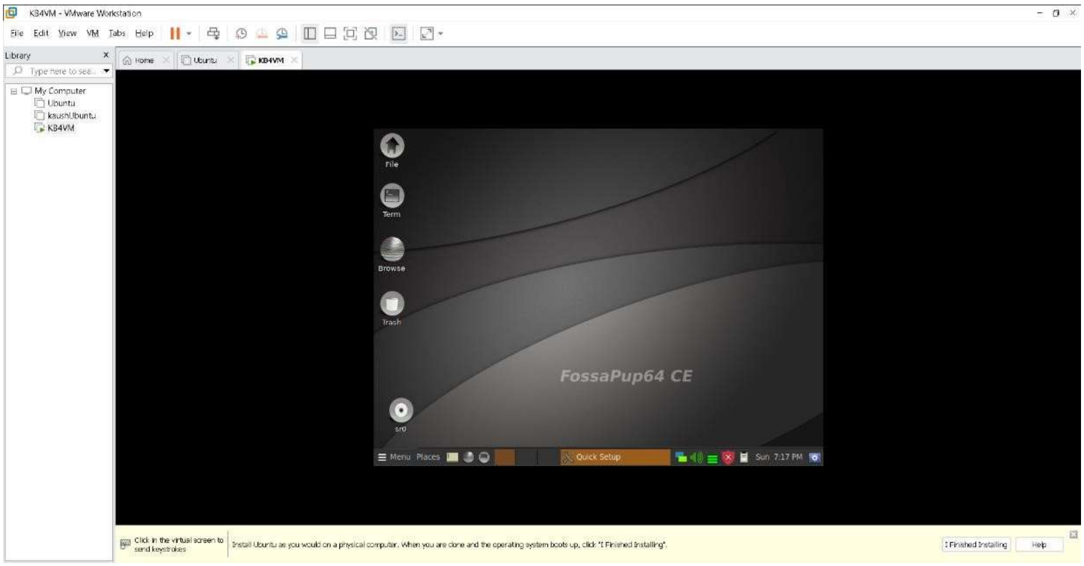


Step 5: Install Operating System:

- Follow the on-screen instructions to install the operating system from the ISO image.
- Complete the installation process by setting up the user account, timezone, and other preferences.



- Now your VM is ready to use.



PRACTICAL :5

AIM: A. Desktop Virtualization using VNC

B. Desktop Virtualization using Chrome Remote Desktop.

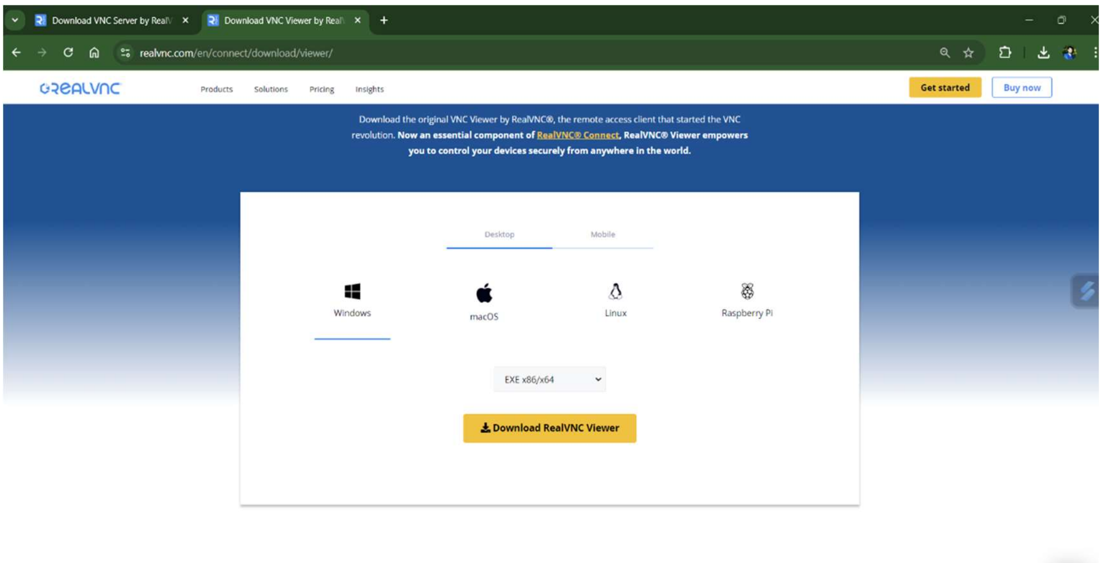
A. Desktop Virtualization Using VNC

Desktop virtualization using VNC (Virtual Network Computing) allows you to remotely control and interact with another computer over a network. VNC enables you to view and control the desktop of a remote computer (the host machine) from another device (the client machine), as if you were sitting in front of it. This is especially useful for tasks such as remote support, accessing files and applications from anywhere, and managing multiple systems.

To achieve this, you need to set up a VNC Server on the host machine, which will be accessed by the VNC Viewer on the client machine.

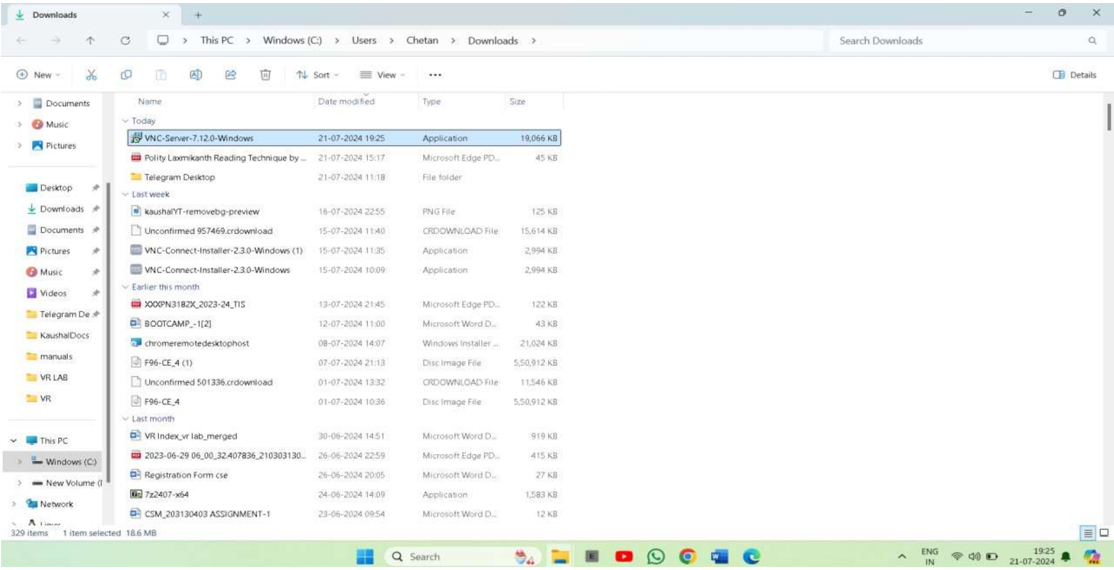
Step 1: Install VNC Server on the Host Machine

1. **Download and Install VNC Server:**
   - Go to the VNC Connect website and download the VNC Server for your operating system.

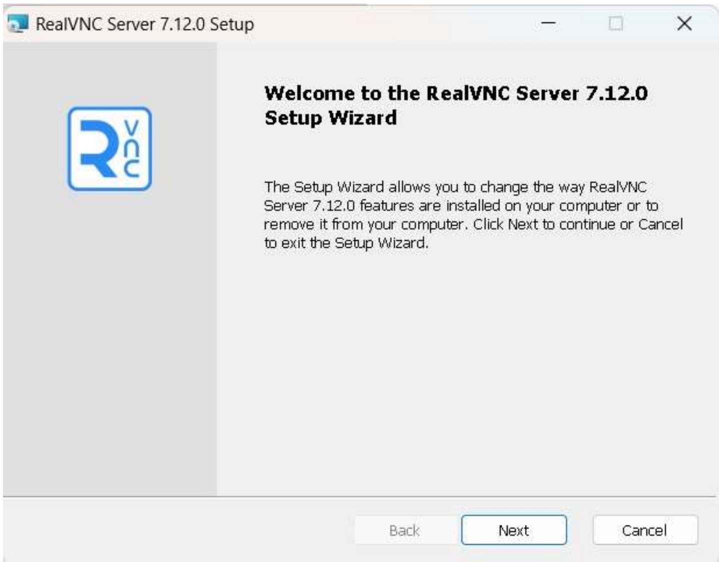


2. **Run the Installer:**

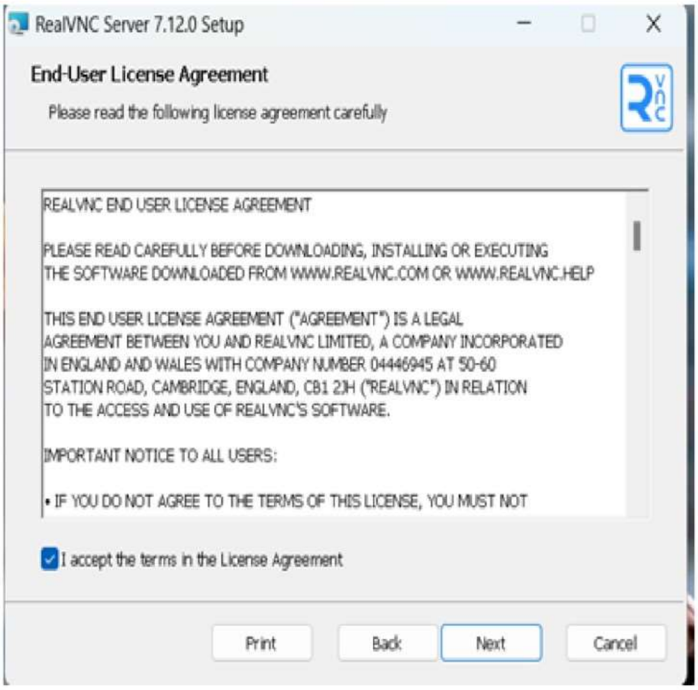
- Locate the downloaded installer file and double-click it to run the installer.



- Click 'Next' to start the installation process

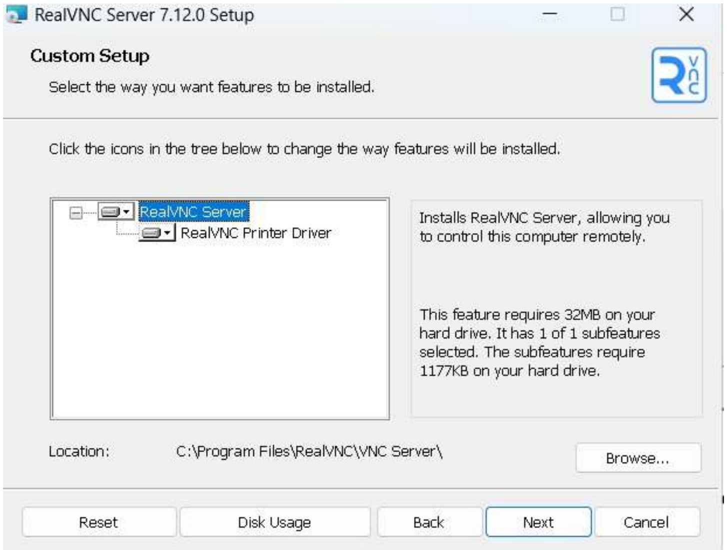


3. **Accept the License Agreement:**
   - Read the license agreement.
   - Check the box to accept the terms of the license agreement.
   - Click 'Next'



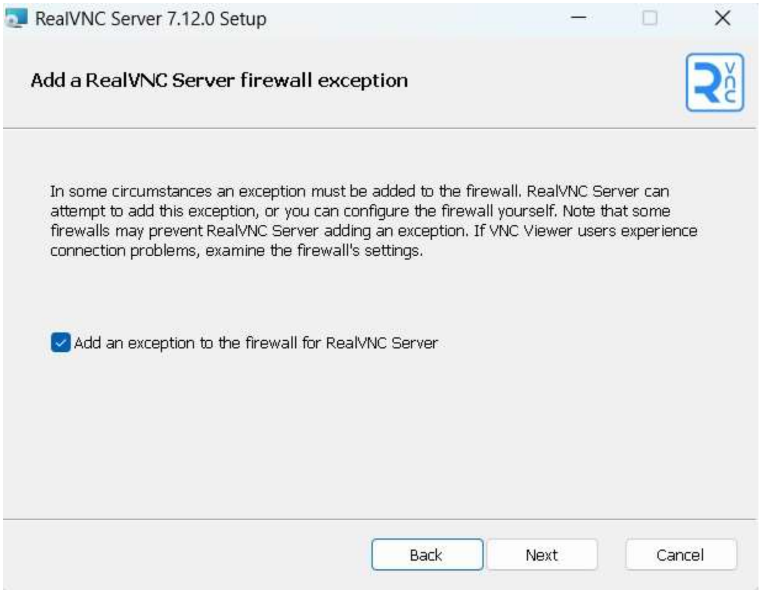
4. **Choose Installation Location:**

 Choose the destination folder for the installation or leave it at the default location and click Next.

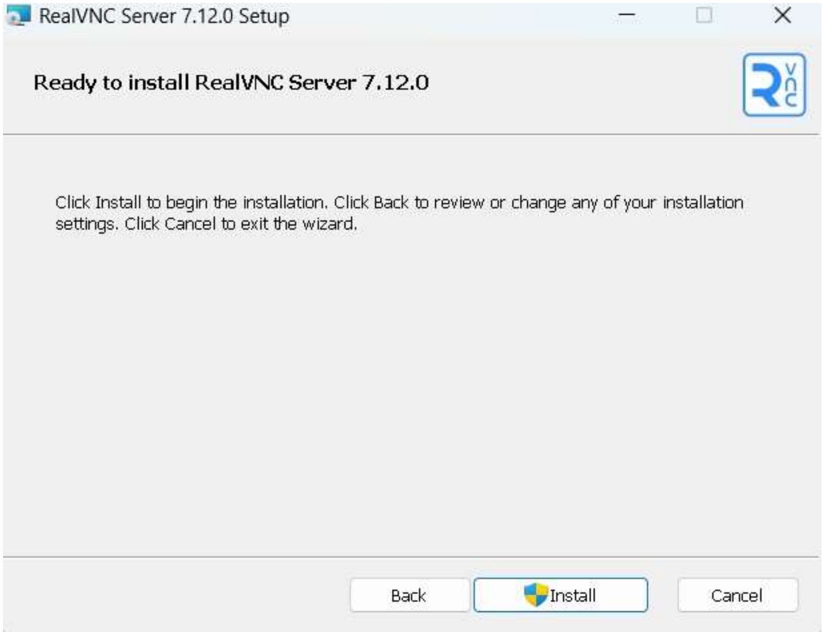


5. **Add VNC Server Firewall Exception:**

- You will see an option to "AddVNC Server to the system firewall exceptions."
- Check the box to allow this and click 'Next'

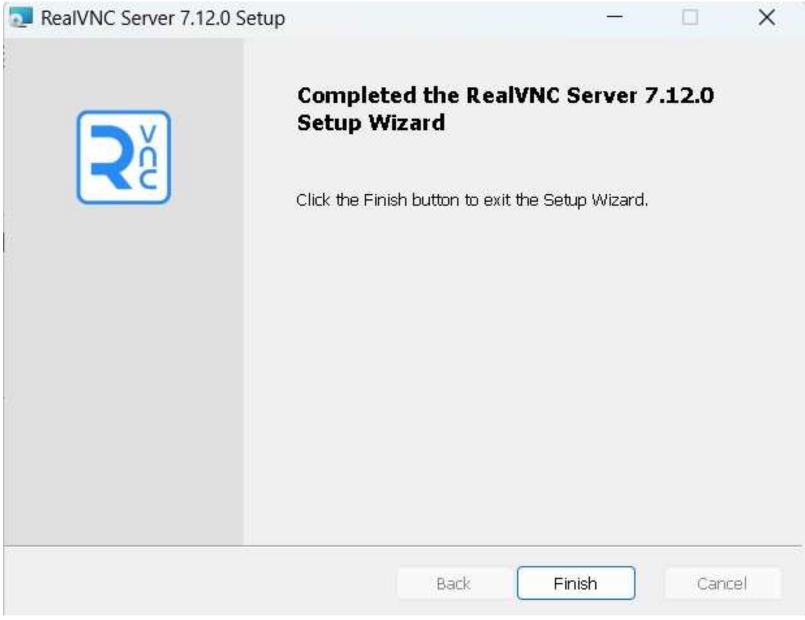


6. **Ready to Install:**
   - Review your installation settings.
   - Click 'Install' to begin the installation.

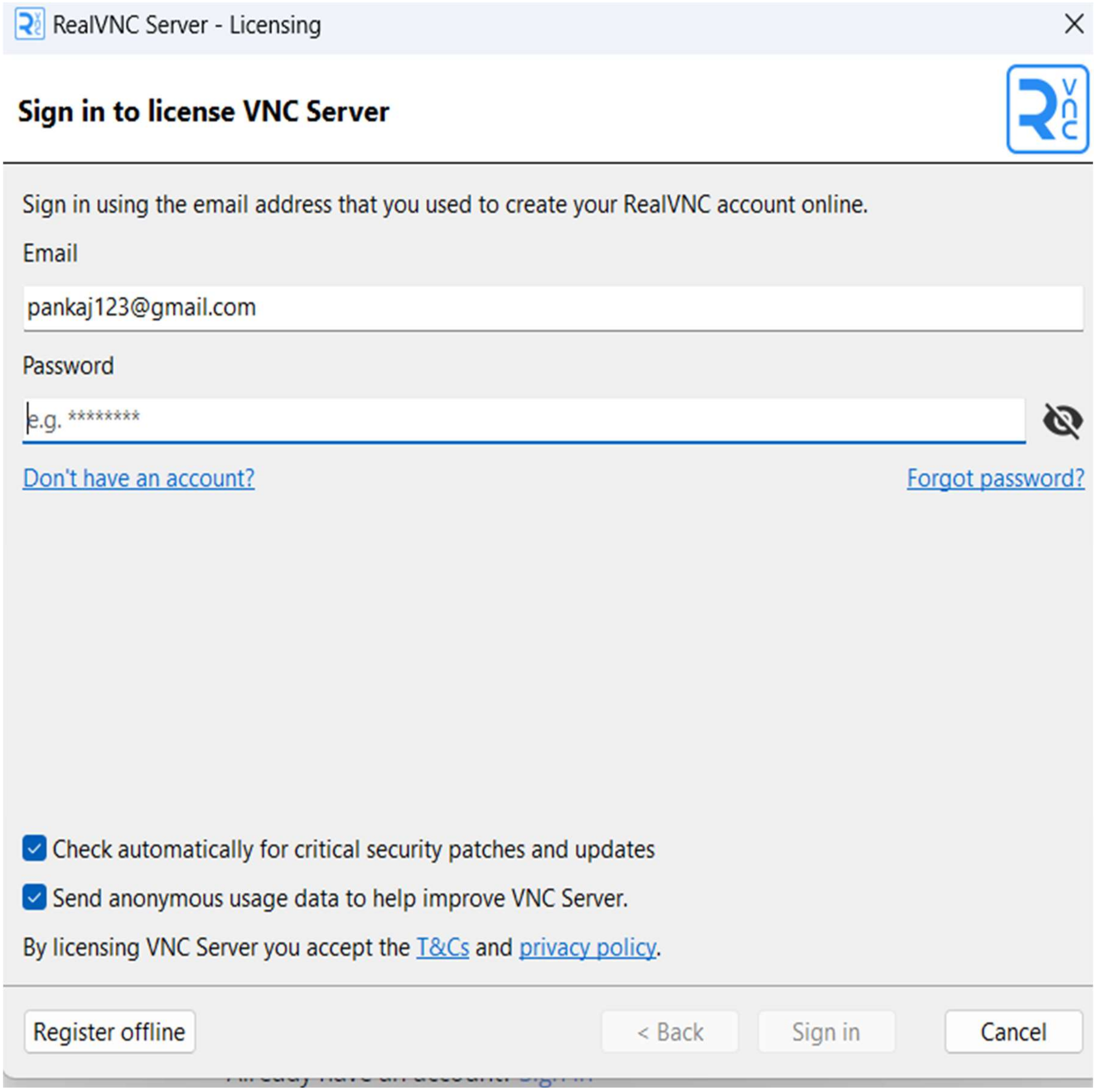


7. **Complete Installation:**

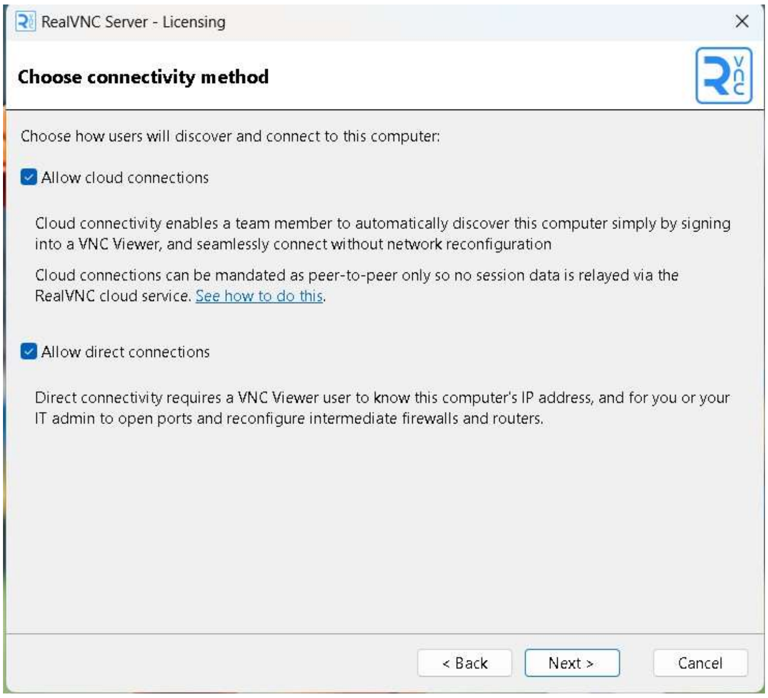
 Once the installation is complete, click 'Finish'.

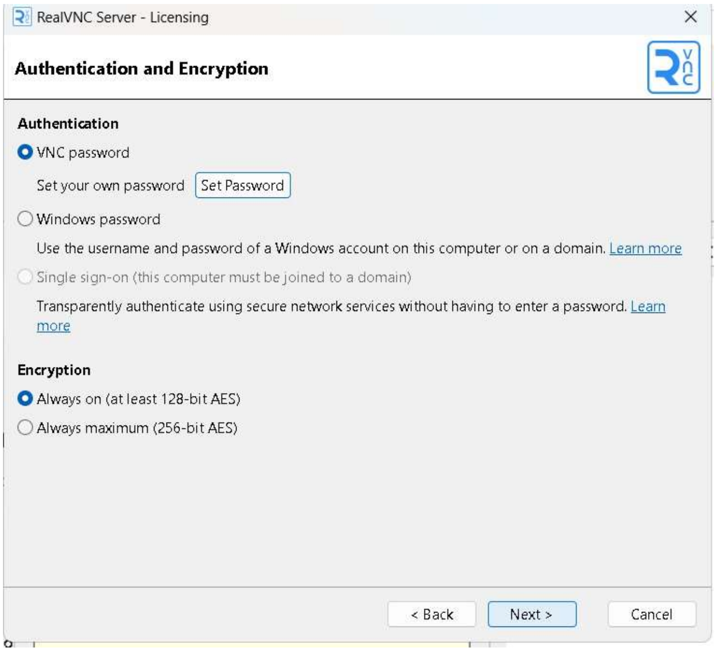


8. **Sign In to License VNC Server:**
   - VNC Server will prompt you to sign in to license the server.
   - Enter your VNC Connect account credentials and click 'Sign In'.



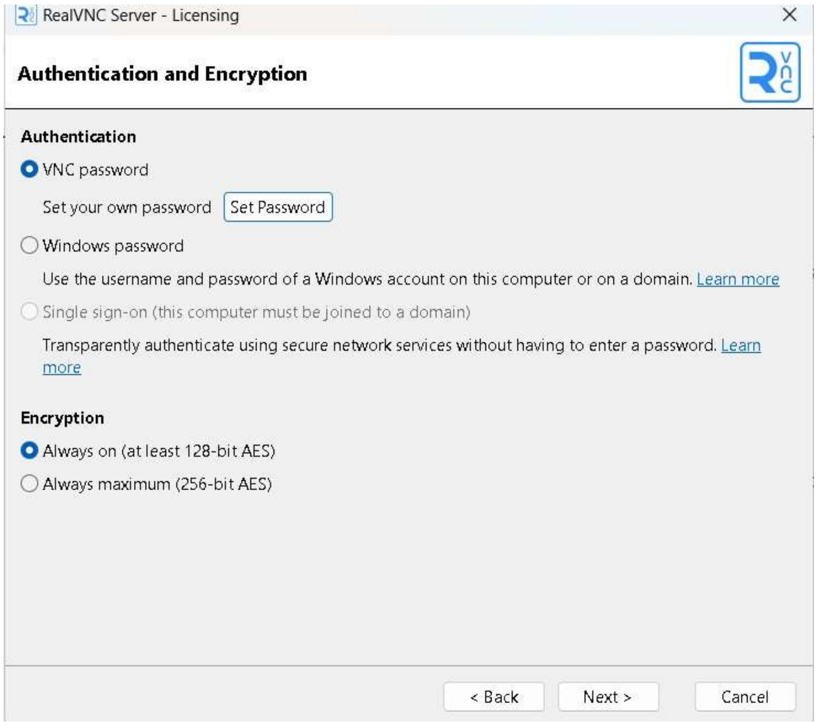
9. **Configure Connectivity:**
    - Choose your connectivity method (e.g., cloud or direct).
    - Click 'Next'.





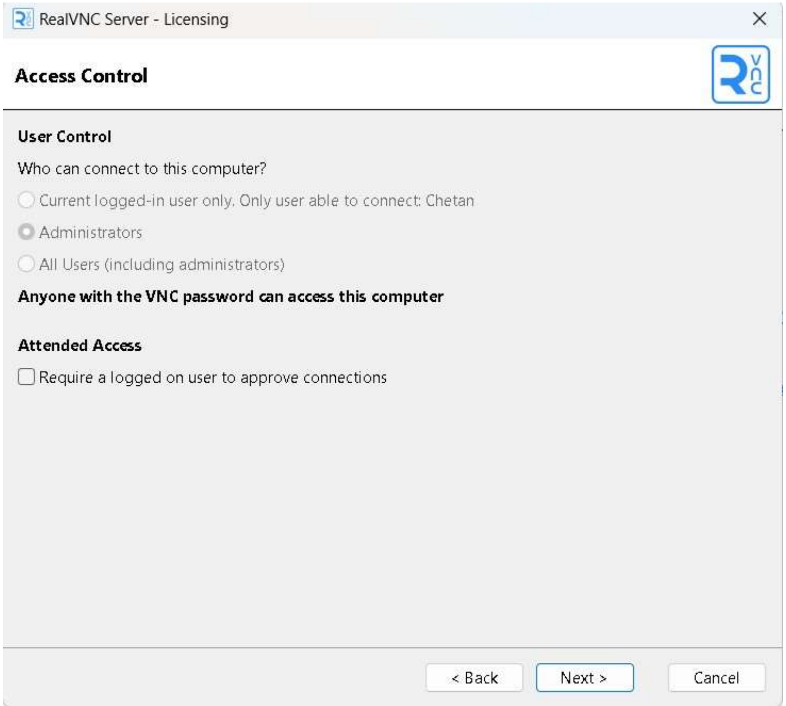
10. **Configure Authentication and Encryption:**

- Set up your preferred authentication method and encryption level.



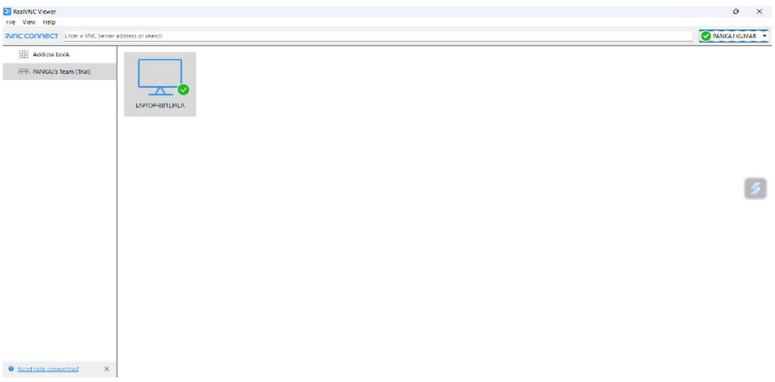
- Set Password for Remote Access Set a password for remote access.
- Click 'OK' to confirm the settings.
- Click 'Next'.

11. **Configure Access Control:**
    - Configure the access control settings according to your preferences.
    - Click 'Next'.



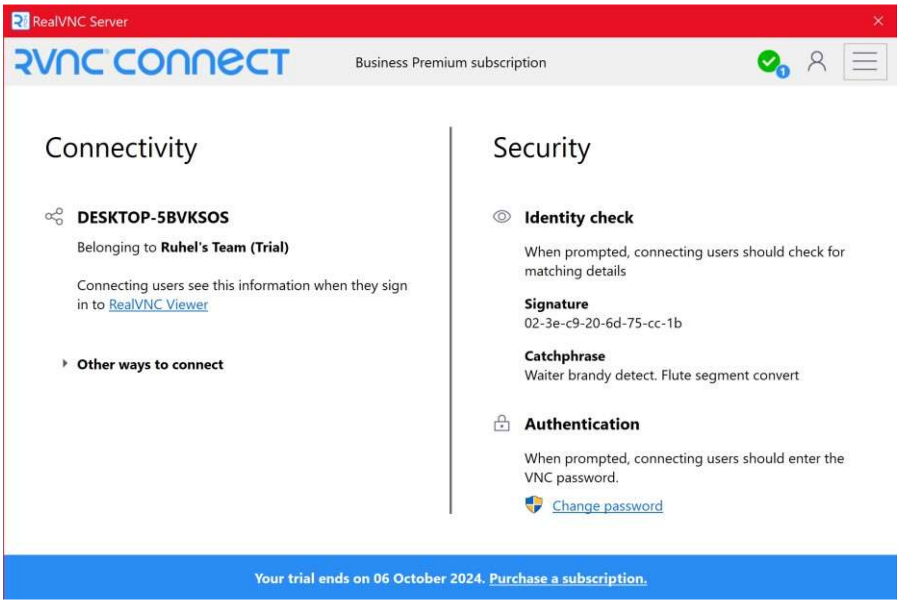
12. **Review Summary:**

- Review the summary of your configuration settings.



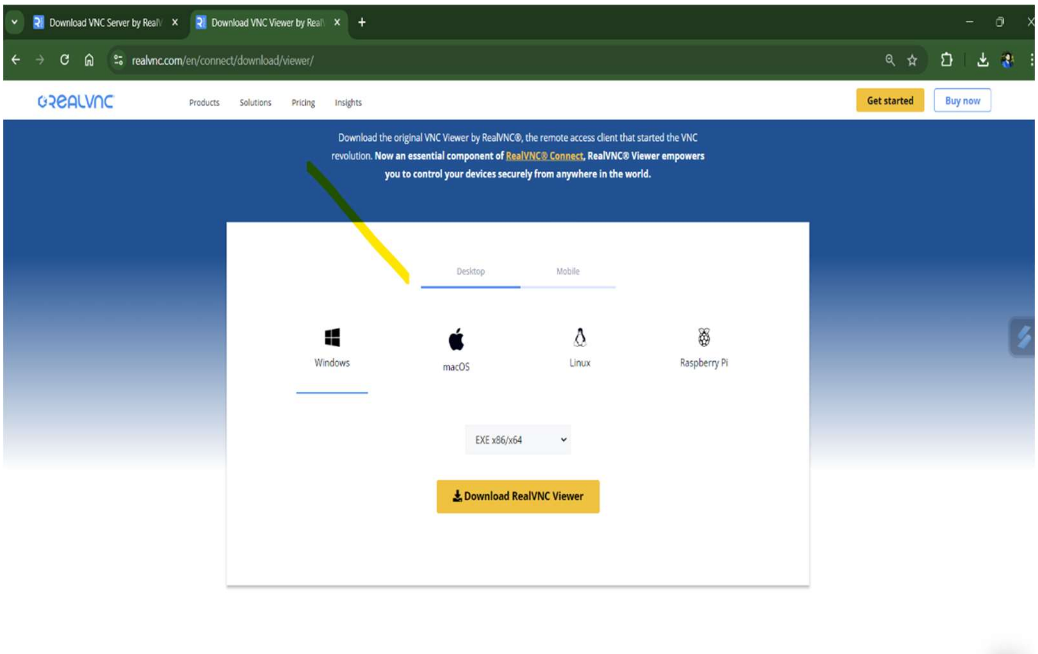
- Click 'Apply' to complete the setup.

Your VNC Server is now ready and running.



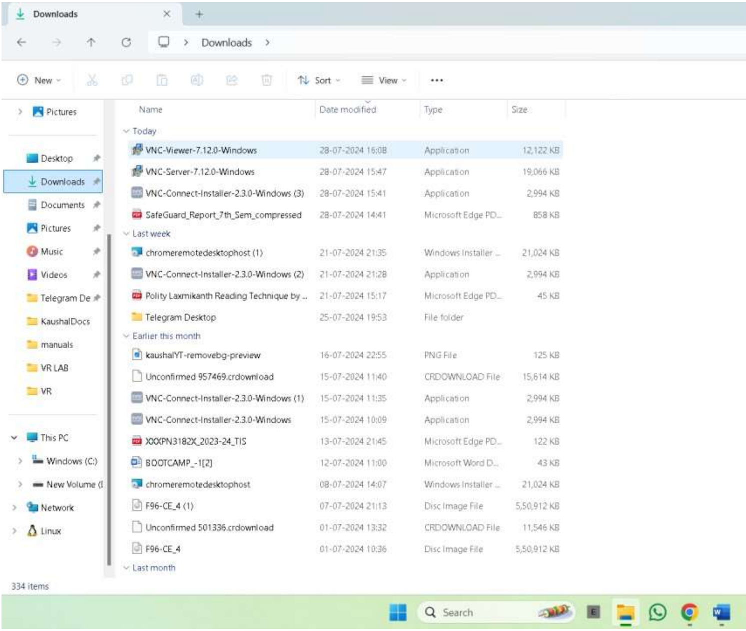
Step 2: Install VNC Viewer on the Client Machine

1. **Download and Install VNC Viewer:**
   - Go to the VNC Connect website and download the VNC Viewer for your operating system.



2. **Run the Installer:**

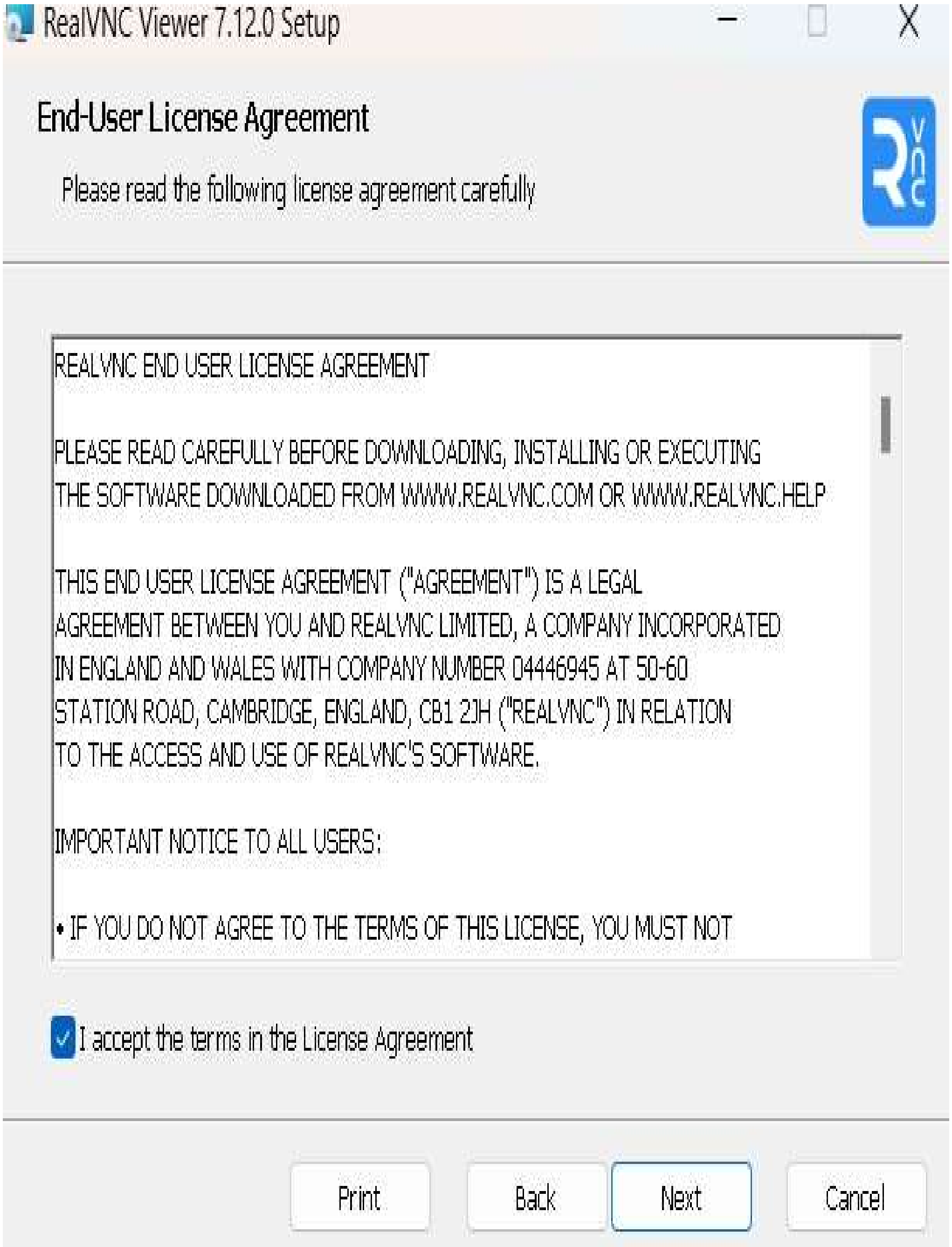
- Locate the downloaded installer file and double-click it to run the installer.



- Click 'Next' to start the installation process

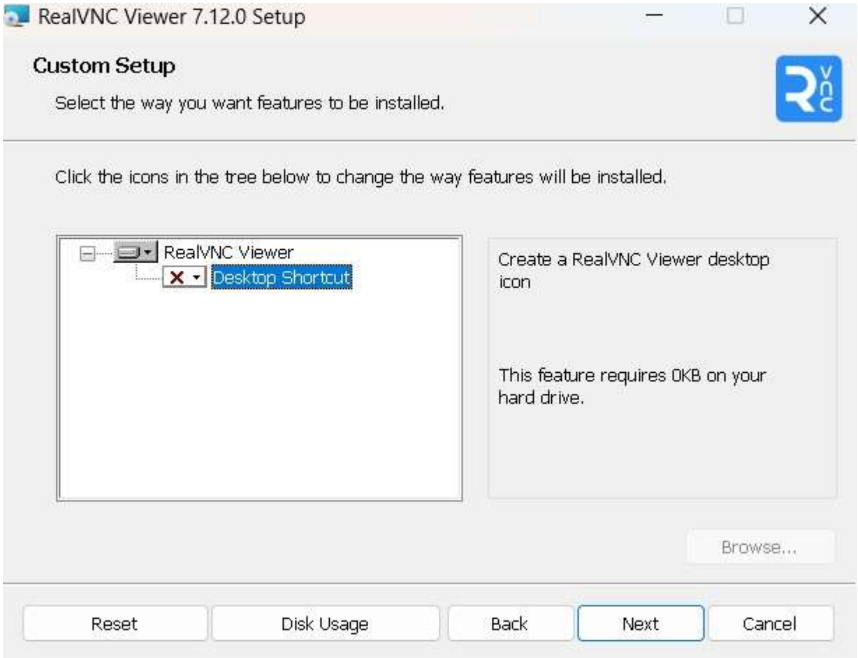
3. **Accept the License Agreement:**

- Read the license agreement.
- Check the box to accept the terms of the license agreement. □ Click 'Next'.



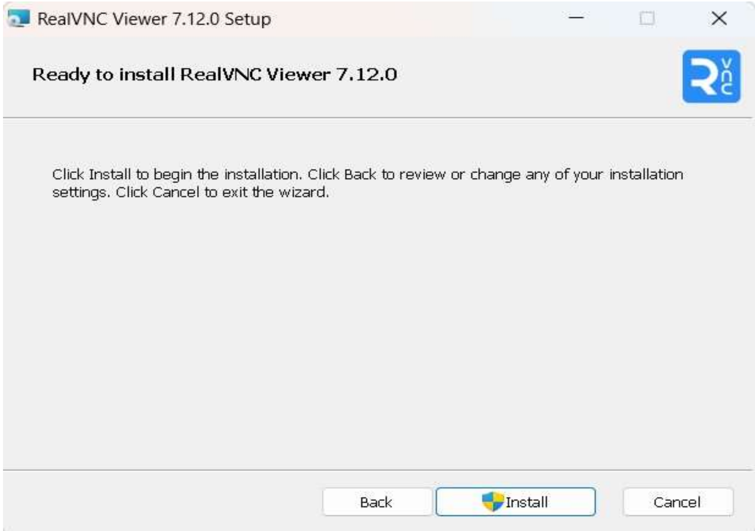
4. **Choose Installation Location:**

- Choose the destination folder for the installation or leave it at the default location.
- Click 'Next'.



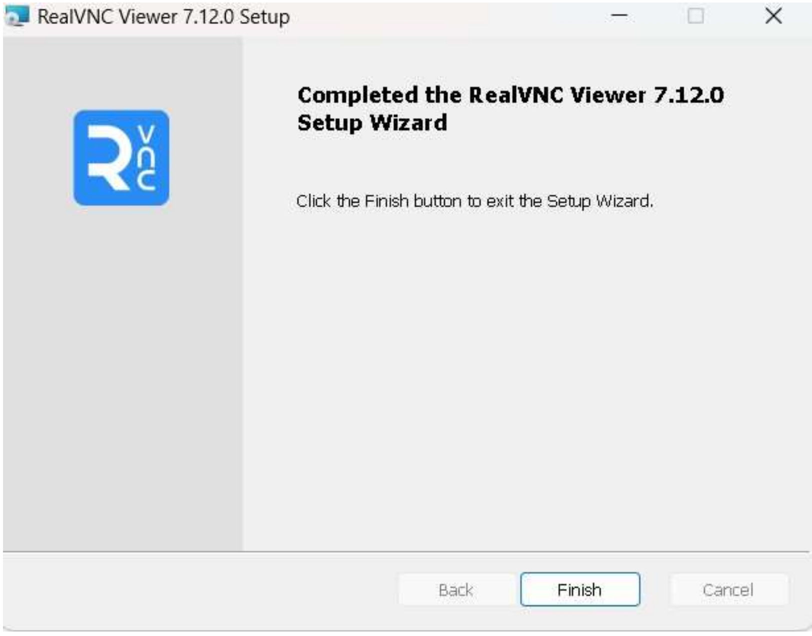
5. **Ready to Install:**

- Review your installation settings.
- Click 'Install' to begin the installation.



6. **Complete Installation:**

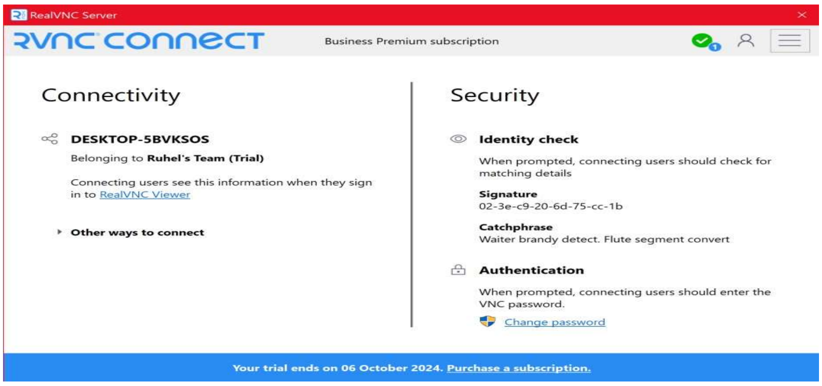
 Once the installation is complete, click 'Finish'.



Step 3: Accessing Host Machine via Client Machine

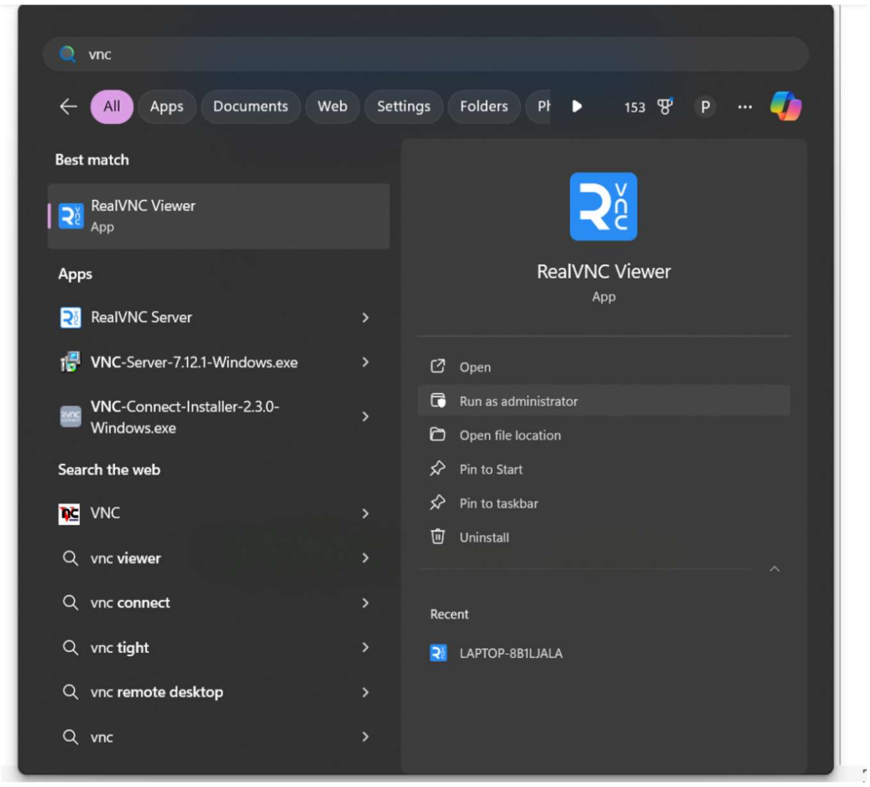
1. **Ensure VNC Server is Running:**

 Make sure that the VNC Server application is open and running on the host machine.

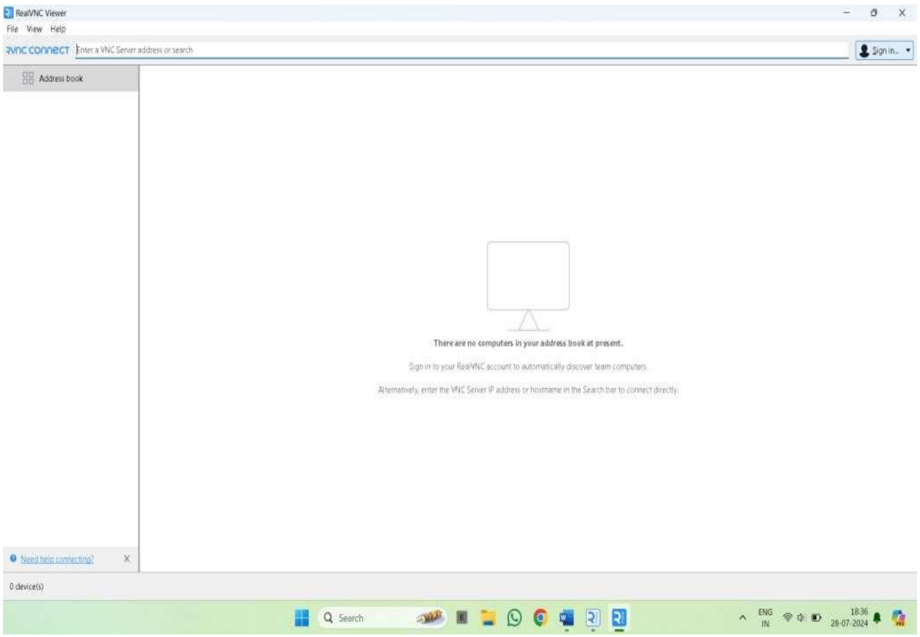


2. **Sign In to VNC Viewer:**

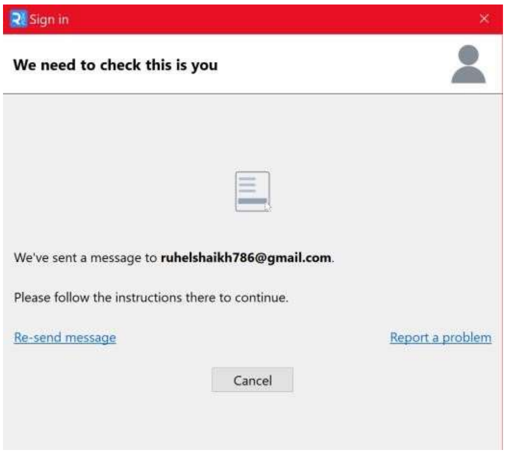
- Open VNC Viewer on the client machine.



- In the top-right corner of VNC Viewer, click 'Sign In'.



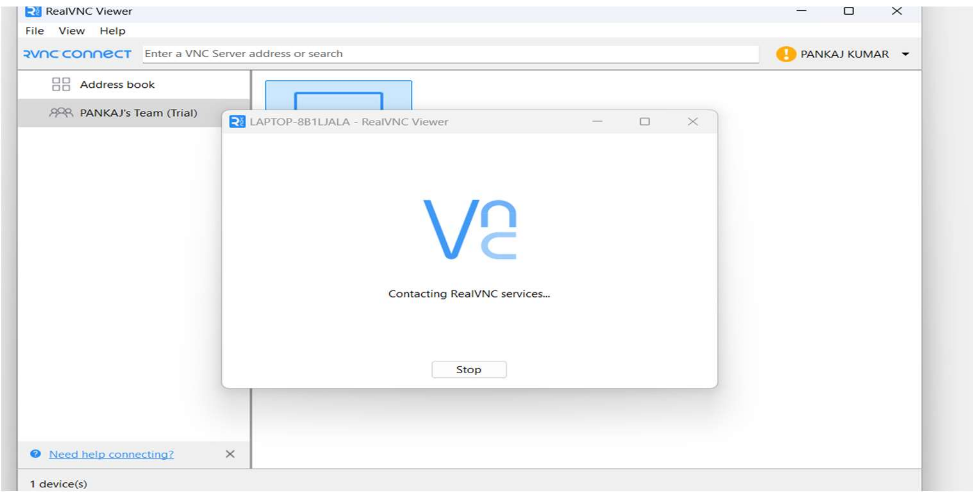
- Enter the email address and password you used for VNC Server.



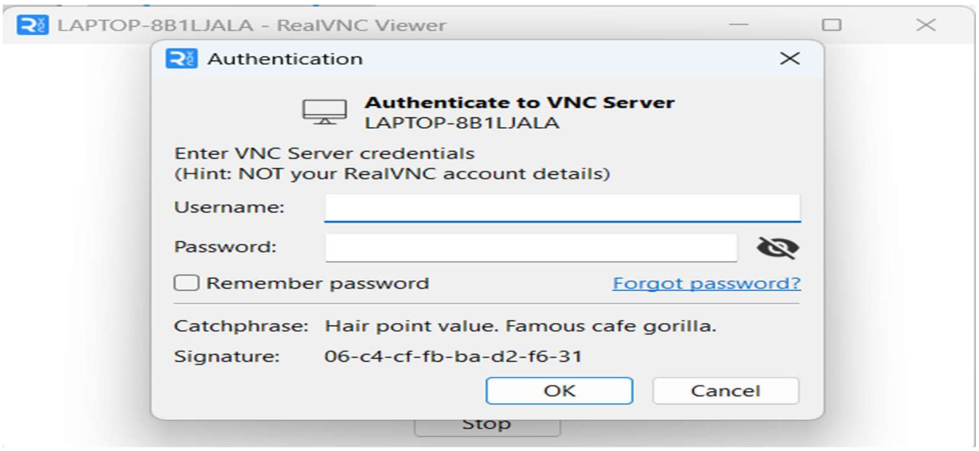
- Click 'Sign In'.

3. **Select the VNC Server:**
   - After signing in, you will see a list of available devices.

   - Click on your device (e.g., "PANKAJ KUMAR").

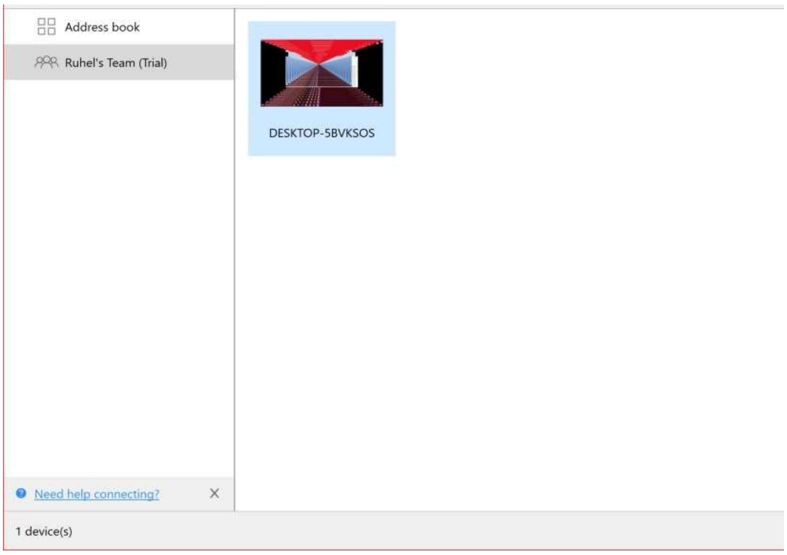


4. **Authenticate Connection:**
    - You will be prompted to enter the password you set for the VNC Server.



    - Enter the password and click 'OK'.

5. **Remote Access:**
    - Once authenticated, you will see the desktop of your host machine on your client machine.
    - You can now operate your host machine remotely from the client machine

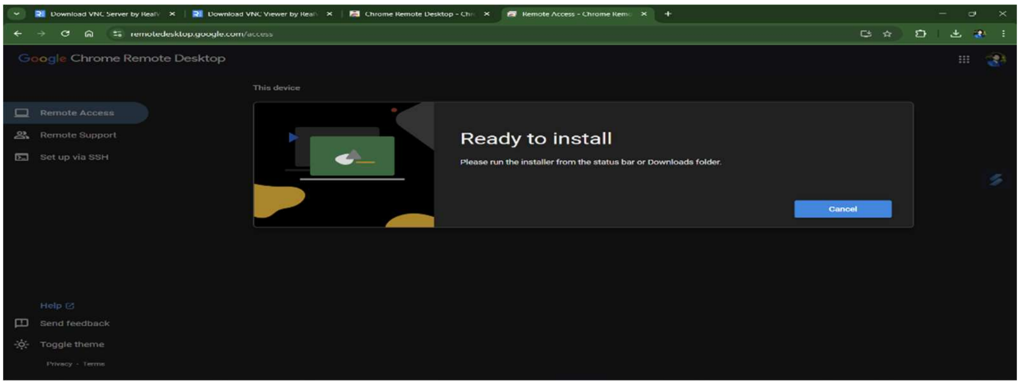


B. Desktop Virtualization Using Chrome Remote Desktop

Desktop virtualization involves hosting desktop operating systems on a central server, allowing users to access their desktops remotely from any device with an internet connection. While Chrome Remote Desktop is primarily designed for remote access to a physical machine, it can be adapted for a basic form of desktop virtualization with certain limitations.

Install Chrome Remote Desktop Extension:

- Head to the Chrome Web Store - Chrome Remote Desktop and click on “Add to Chrome” to install the Chrome Remote Desktop extension into your Chrome browser.



Open the Extension:

- Once installed, click on the Chrome Remote Desktop extension icon in the top right corner of your Chrome browser.

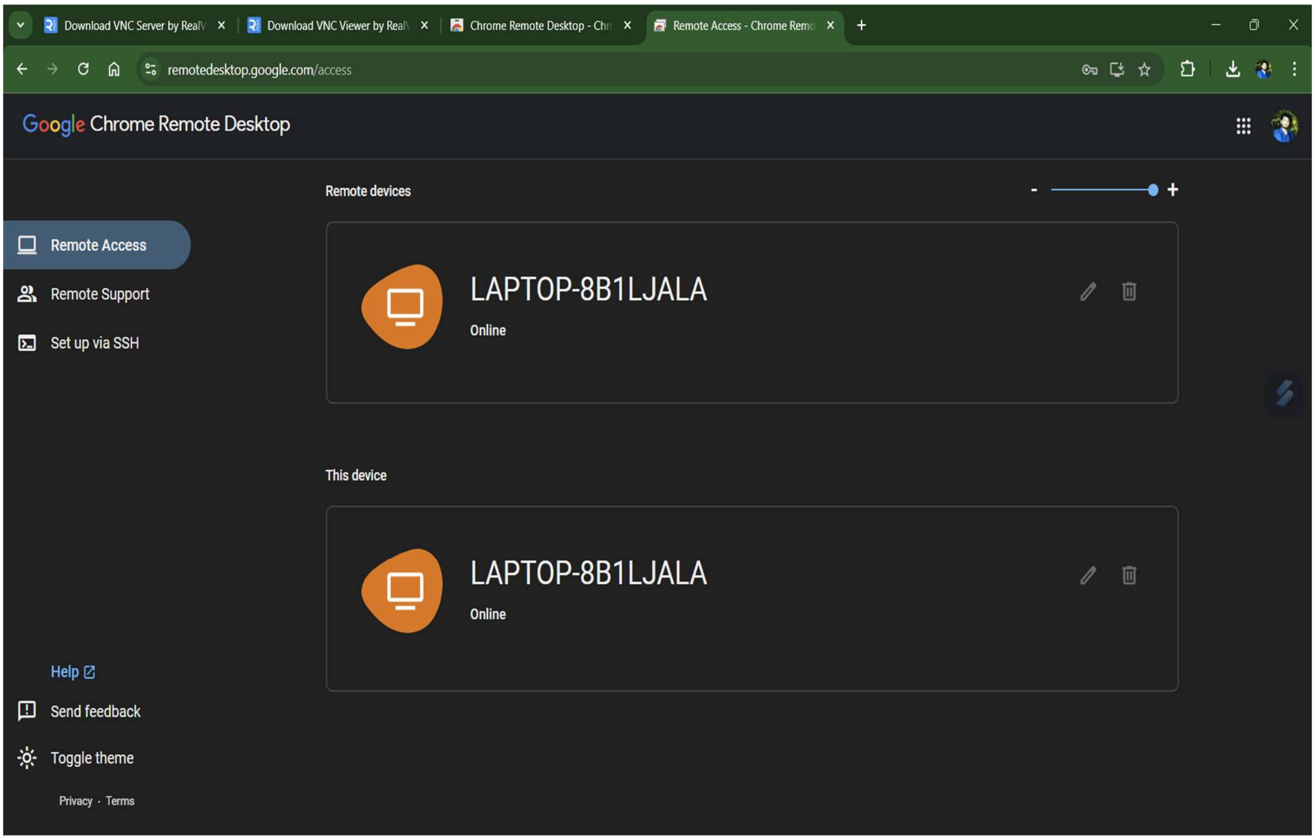
Enable Remote Access:

- In the extension window, click on “Remote Access” in the left-hand menu.

- Click on “Turn on” to enable remote access.

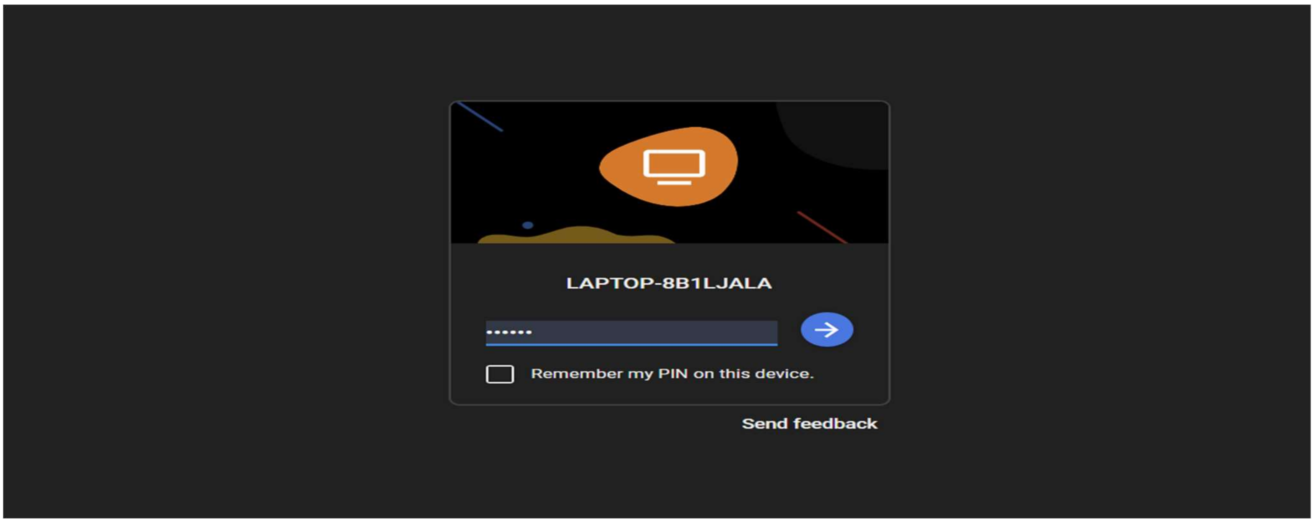
Set Up Your Device:

- Enter a name for your device in the “Enter a name” field.
- Click “Next.”



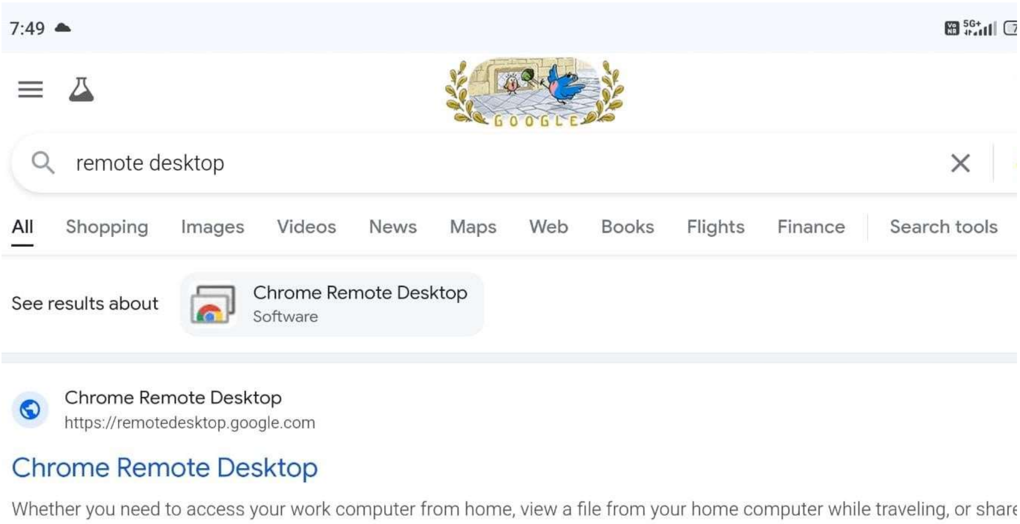
Set Up a PIN for Authentication:

- Create and enter a six-digit PIN number for authentication in the “Create a PIN” field.
- Confirm your PIN by re-entering it.
- Click “Start” to complete the setup on this device.



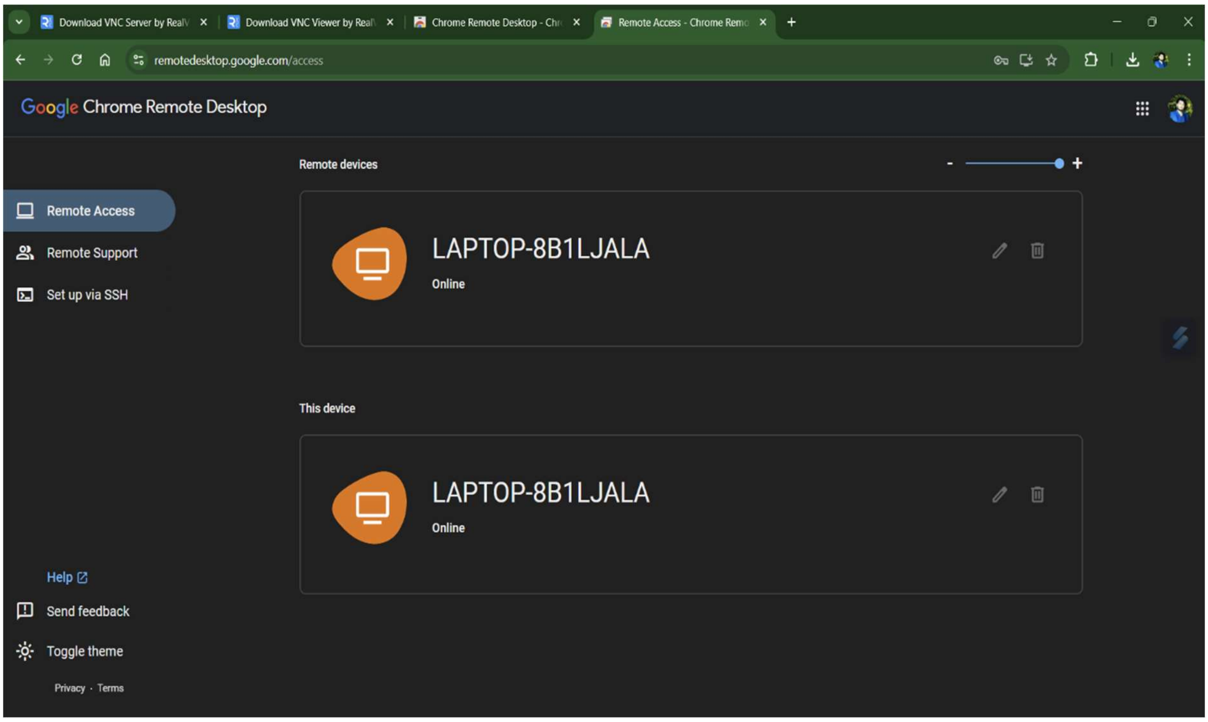
Set Up on Another Device:

- On another device (e.g., laptop or mobile), open Google Chrome and sign in with the same Gmail ID used in the previous steps.
- Go to the Chrome Remote Desktop website.

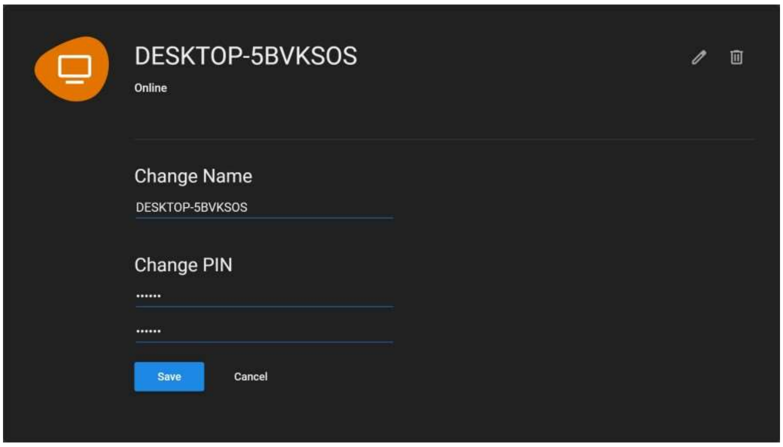


Access Your Remote Device:

- You will see the device name you set earlier listed under “Remote Devices.” ▯ Click on your device name.



- Enter the six-digit PIN you set up previously.



- You can now control your PC remotely from this second device.